); DROP TABLE Teams;–

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Contest (1)

template.cpp

21 lines

```
#include <bits/stdc++.h>

using namespace std;
using ll = long long;

#define FOR(i, a, b) for (int i = a; i < (b); i++)
#define F0R(i, b) FOR(i, 0, b)
#define all(x) begin(x), end(x)
#define vt vector
#define size(x) ((int) (x).size())
#define ROF(i, a, b) for (int i = (b) - 1; i >= (a); i--)
#define pb push_back
#define f first
#define s second
using vi = vt<int>;

int main() {
    cin.tie(0)->sync_with_stdio(0);
    cin.exceptions(cin.failbit);

}
```

hash.sh

5 lines

```
# Hashes a file, ignoring all whitespace and comments. Use for
# verifying that code was correctly typed.
cpp -dD -P -fpreprocessed | tr -d '[:space:]'| md5sum |cut -c-6
# eg:
# cat sus.cpp | cpp -dD -P -fpreprocessed | tr -d '[:space:]'| md5sum
     |cut -c-6
```

troubleshoot.txt

69 lines

```
Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
Make sure to submit the right file.

Wrong answer:
Print your solution! Print debug output, as well.
Are you clearing all data structures between test cases?
Can your algorithm handle the whole range of input?
Read the full problem statement again.
Do you handle all corner cases correctly?
Have you understood the problem correctly?
Any uninitialized variables?
Any overflows?
Confusing N and M, i and j, etc.?
Are you sure your algorithm works?
What special cases have you not thought of?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some testcases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
Is your output format correct? (including whitespace)
Rewrite your solution from the start or let a teammate do it.
How sure are you its a precision error and not a logic error?

Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Any possible division by 0? (mod 0 for example)
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:
Do you have any possible infinite loops?
What is the complexity of your algorithm?
Are you copying a lot of unnecessary data? (References)
How big is the input and output? (consider scanf)
Avoid vector, map. (use arrays/unordered_map)
What do your teammates think about your algorithm?

Memory limit exceeded:
What is the max amount of memory your algorithm should need?
Are you clearing all data structures between test cases?

Stupid bugs:
Did you abs() and type your epsilons correctly?
Is your maximum precomputed value really the maximum?
Is your sorting comparator a strictly weak ordering?
Are your binary search bounds big enough?
Are you reading into a char when it should be a string?
Are your return types big enough?
Are you missing arguments and accidentally defaulting them?
Are you shadowing variables?
Is your Pi correct (atan2 is y, x)

Last ditch attempts:
#define int ll
#define int lll
#define sort stable_sort
```

stress.sh

11 lines

```
# A and B are executables you want to compare, gen takes int
# as command line arg. Usage: 'sh stress.sh'
for((i = 1; ; ++i)); do
    echo $i
    ./gen $i > int
    ./A < int > out1
    ./B < int > out2
    diff -w out1 out2 || break
    # diff -w <(./A < int) <(./B < int) || break
    # ./A < in > out || break
done
```

Mathematics (2)

2.1 Equations

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The extremum is given by $x = -b/2a$.

$$\begin{aligned} ax + by &= e \\ cx + dy &= f \end{aligned} \Rightarrow \begin{aligned} x &= \frac{ed - bf}{ad - bc} \\ y &= \frac{af - ec}{ad - bc} \end{aligned}$$

In general, given an equation $Ax = b$, the solution to a variable x_i is given by

$$x_i = \frac{\det A_i'}{\det A}$$

where A_i' is A with the i'th column replaced by b.

2.2 Recurrences

If $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$, and $r_1, \ldots, r_k$ are distinct roots of $x^k - c_1 x^{k-1} - \cdots - c_k$, there are $d_1, \ldots, d_k$ s.t.

$$a_n = d_1 r_1^n + \cdots + d_k r_k^n.$$

Non-distinct roots r become polynomial factors, e.g. $a_n = (d_1 n + d_2) r^n$.

2.3 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$
$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$

$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$(V + W) \tan(v - w)/2 = (V - W) \tan(v + w)/2$$

where V, W are lengths of sides opposite angles v, w.

$$a\cos x + b\sin x = r\cos(x-\phi)$$
$$a\sin x + b\cos x = r\sin(x+\phi)$$

where $r = \sqrt{a^2+b^2}, \phi = \operatorname{atan2}(b,a)$.

2.4 Geometry

2.4.1 Triangles

Side lengths: a, b, c

Semiperimeter: $p = \dfrac{a+b+c}{2}$

Area: $A = \sqrt{p(p-a)(p-b)(p-c)}$

Circumradius: $R = \dfrac{abc}{4A}$

Inradius: $r = \dfrac{A}{p}$

Length of median (divides triangle into two equal-area triangles): $m_a = \frac{1}{2}\sqrt{2b^2+2c^2-a^2}$

Length of bisector (divides angles in two):

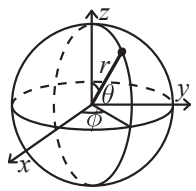
$$s_a = \sqrt{bc\left[1-\left(\frac{a}{b+c}\right)^2\right]}$$

Law of sines: $\dfrac{\sin\alpha}{a} = \dfrac{\sin\beta}{b} = \dfrac{\sin\gamma}{c} = \dfrac{1}{2R}$

Law of cosines: $a^2 = b^2 + c^2 - 2bc\cos\alpha$

Law of tangents: $\dfrac{a+b}{a-b} = \dfrac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}$

2.4.2 Spherical coordinates



$$x = r\sin\theta\cos\phi \qquad r = \sqrt{x^2+y^2+z^2}$$
$$y = r\sin\theta\sin\phi \qquad \theta = \operatorname{acos}(z/\sqrt{x^2+y^2+z^2})$$
$$z = r\cos\theta \qquad \phi = \operatorname{atan2}(y,x)$$

2.5 Derivatives/Integrals

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\tan x = 1+\tan^2 x \qquad \frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$
$$\int \tan ax = -\frac{\ln|\cos ax|}{a} \qquad \int x\sin ax = \frac{\sin ax - ax\cos ax}{a^2}$$
$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) \qquad \int xe^{ax}dx = \frac{e^{ax}}{a^2}(ax-1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.6 Sums

$$c^a + c^{a+1} + \cdots + c^b = \frac{c^{b+1}-c^a}{c-1}, c \neq 1$$

$$1+2+3+\cdots+n = \frac{n(n+1)}{2}$$
$$1^2+2^2+3^2+\cdots+n^2 = \frac{n(2n+1)(n+1)}{6}$$
$$1^3+2^3+3^3+\cdots+n^3 = \frac{n^2(n+1)^2}{4}$$
$$1^4+2^4+3^4+\cdots+n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

2.7 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots, (-\infty < x < \infty)$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots, (-1 < x \leq 1)$$
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \ldots, (-1 \leq x \leq 1)$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots, (-\infty < x < \infty)$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots, (-\infty < x < \infty)$$

2.8 Probability theory

Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x. It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x xp_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x-\mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation. If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX+bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y,

$$V(aX+bY) = a^2V(X) + b^2V(Y).$$

Data structures (3)

OrderStatisticTree.h

Description: A set (not multiset!) with support for finding the n'th element, and finding the index of an element. To get a map, change `null_type`.
Time: $\mathcal{O}(\log N)$

782797, 17 lines

```
#include <bits/extc++.h>
using namespace __gnu_pbds;

// MEGA WARNING: use A.swap(B) instead of swap(A, B)!
template<class T>
using Tree = tree<T, null_type, less<T>, rb_tree_tag,
  tree_order_statistics_node_update>;

void example() {
  Tree<int> t, t2; t.insert(8);
  auto it = t.insert(10).first;
  assert(it == t.lower_bound(9));
  assert(t.order_of_key(10) == 1);
  assert(t.order_of_key(11) == 2);
  assert(*t.find_by_order(0) == 8);
  t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

MonotonicMap.h

Description: A data structure for performing prefix/suffix min/max queries with insertions. Useful replacement for sparse segtrees.
Time: $\mathcal{O}(\log N)$

1db6eb, 14 lines

```
// dir = less<> for suffix queries, greater<> for prefix
// cmp = less_equal<> for min, greater_equal<> for max
template<class dir, class cmp>
struct RangeQuery {
  map<int, ll, dir> data;
  void ins(int k, ll v) {
    if (auto it = data.lower_bound(k); it != data.end() && cmp{}(it->s
        , v)) return;
    auto it = data.insert_or_assign(k, v).f;
    while (it != data.begin() && cmp{}(v, prev(it)->s)) data.erase(
        prev(it));
  }
  ll query(int k) { // inclusive: careful UB, insert infinity value
    return data.lower_bound(k)->s;
  }
};
```

HashMap.h

Description: Hash map with mostly the same API as unordered_map, but ~3x faster. Uses 1.5x memory. Initial capacity must be a power of 2 (if provided).

d77092, 7 lines

```
#include <bits/extc++.h>
// To use most bits rather than just the lowest ones:
struct chash { // large odd number for C
  const uint64_t C = ll(4e18 * acos(0)) | 71;
  ll operator()(ll x) const { return __builtin_bswap64(x*C); }
};
__gnu_pbds::gp_hash_table<ll, int, chash> h({}, {}, {}, {}, {1 << 16})
    ;
```

SegmentTree.h

Description: Generic segment tree for associative operations. `id` is for the identity object.
Time: $\mathcal{O}(\log N)$

f519dc, 20 lines

```
struct Segtree {
  int n;
  vt<T> seg;
  void init(int _n) {
    for (n = 1; n < _n; n *= 2);
    seg.resize(2 * n, id);
  }
  void upd(int i, T v) {
    seg[i += n] = v;
    while (i /= 2) seg[i] = seg[2 * i] + seg[2 * i + 1];
  }
  T query(int l, int r) {
    T lhs = id, rhs = id;
    for (l += n, r += n; l < r; l /= 2, r /= 2) {
      if (l & 1)) lhs = lhs + seg[l++];
      if (r & 1) rhs = seg[--r] + rhs;
    }
    return lhs + rhs;
  }
};
```

LazySegtree.h

Description: Generic lazy segment tree
Usage: `Implement +, upd and id for Node object; += and id for Lazy object.`
Time: $\mathcal{O}(\log N)$.

c33fda, 65 lines

```
struct Lazy {
  ll v;
  bool inc;
  void operator+=(const Lazy &b) {
    if (b.inc) v += b.v;
    else v = b.v, inc = false;
  }
};

struct Node {
  ll mx, sum;
  Node operator+(const Node &b) {
    return {max(mx, b.mx), sum + b.sum};
  }
  void upd(const Lazy &u, int l, int r) {
    if (!u.inc) mx = sum = 0;
    mx += u.v, sum += u.v * (r - l);
  }
};

const Lazy lid = {0, true};
const Node nid = {-INF, 0};

const int sz = 1 << 18;
struct Lazyseg {
  vt<Node> seg;
  vt<Lazy> lazy;
  // careful: max is initialised to -1e18; perform range set or write
      build function
  Lazyseg() : seg(2 * sz, nid), lazy(2 * sz, lid) {}
  void push(int i, int l, int r) {
    seg[i].upd(lazy[i], l, r);
    if (r - l > 1) F0R (j, 2) lazy[2 * i + j] += lazy[i];
    lazy[i] = lid;
  }
  void upd(int lo, int hi, Lazy val, int i = 1, int l = 0, int r = sz)
      {
    push(i, l, r);
    if (r <= lo || l >= hi) return;
    if (lo <= l && r <= hi) {
      lazy[i] += val;
      push(i, l, r);
      return;
    }
    int m = (l + r) / 2;
    upd(lo, hi, val, 2 * i, l, m);
    upd(lo, hi, val, 2 * i + 1, m, r);
    seg[i] = seg[2 * i] + seg[2 * i + 1];
  }
  Node query(int lo, int hi, int i = 1, int l = 0, int r = sz) {
    push(i, l, r);
    if (r <= lo || l >= hi) return nid;
    if (lo <= l && r <= hi) return seg[i];
    int m = (l + r) / 2;
    return query(lo, hi, 2 * i, l, m)
      + query(lo, hi, 2 * i + 1, m, r);
  }
  Node& operator[](int i) {
    return seg[i + sz];
  }
  // void pull(int i) {
  //   seg[i] = seg[2 * i] + seg[2 * i + 1];
  // }
  // void build() {
  //   for (int i = n - 1; i > 0; i--) pull(i);
  // }
};
```

SparseSegtree.h

Description: Generic-ish sparse segment tree (point update, range query).
Usage: `Choose appropriate identity element and merge function.`
Time: $\mathcal{O}(\log N)$.

d51c9b, 28 lines

```
using ptr = struct Node*;
const int sz = 1 << 30;
struct Node {
  #define func(a, b) min(a, b)
  #define ID INF
  ll val;
  ptr lc, rc;

  ptr get(ptr& p) { return p ? p : p = new Node {ID}; }

  ll query(int lo, int hi, int l = 0, int r = sz) {
    if (lo >= r || hi <= l) return ID;
    if (lo <= l && r <= hi) return val;
    int m = (l + r) / 2;
    return func(get(lc)->query(lo, hi, l, m),
      get(rc)->query(lo, hi, m, r));
  }

  ll upd(int i, ll nval, int l = 0, int r = sz) {
    if (r - l == 1) return val = nval;
    int m = (l + r) / 2;
    if (i < m) get(lc)->upd(i, nval, l, m);
    else get(rc)->upd(i, nval, m, r);
    return val = func(get(lc)->val, get(rc)->val); // creates extra
  }
  #undef ID
  #undef func
};
```

PersistentSegtree.h

Description: Generic-ish persistent segment tree (point update, range query).
Usage: `Choose appropriate identity element and merge function.`
Time: $\mathcal{O}(\log N)$.

f155de, 30 lines

```
using ptr = struct Node*;
const int sz = 1 << 18;

struct Node {
  #define func(a, b) min(a, b)
  #define ID inf
  int v;
  ptr lc, rc;

  ptr pull(ptr lc, ptr rc) {
    return new Node {func(lc->v, rc->v), lc, rc};
  }

  ptr upd(int i, int nv, int l = 0, int r = sz) {
    if (r - l == 1) return new Node {nv};
    int m = (l + r) / 2;
    if (i < m) return pull(lc->upd(i, nv, l, m), rc);
    else return pull(lc, rc->upd(i, nv, m, r));
  }

  int query(int lo, int hi, int l = 0, int r = sz) {
    if (lo >= r || hi <= l) return ID;
    if (lo <= l && r <= hi) return v;
    int m = (l + r) / 2;
    return func(lc->query(lo, hi, l, m),
      rc->query(lo, hi, m, r));
  }
  #undef id
  #undef func
};
```

LiChaoTree.h

Description: LiChao tree
Usage: `self explanatory i think`
Time: $\mathcal{O}(\log N)$.

00d540, 36 lines

```
struct Line {
  ll m, c;
  ll operator()(ll x) {
    return m * x + c;
  }
};

const ll sz = 1ll << 30;

using ptr = struct Node*;
struct Node {
  ptr lc, rc;
  Line line;

  Node(Line _line) {
    line = _line;
    lc = rc = 0;
  }
};

// min tree (flip signs for max)
void add(ptr& n, Line loser, ll l = 0, ll r = sz) {
  if (n ? 0 : n = new Node(loser)) return;
  ll m = (l + r) / 2;
  if (loser(m) < n->line(m)) swap(loser, n->line);
  if (r - l == 1) return;
  if (loser(l) < n->line(l)) add(n->lc, loser, l, m);
  else add(n->rc, loser, m, r);
}

ll query(ptr n, ll x, ll l = 0, ll r = sz) {
  if (!n) return sz;
```

```
  ll m = (l + r) / 2;
  if (x < m) return min(n->line(x), query(n->lc, x, l, m));
  else return min(n->line(x), query(n->rc, x, m, r));
}
```

FunnySparseTable.h

Description: Generic sparse table for idempotent operations.
Usage: Define the desired operation
Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(1)$ query.

adc2cc, 15 lines

```
template<class T> struct RMQ {
  vt<vt<T>> dp;
  T query(int l, int r) {
    int d = max(0, __lg(r - l - 1));
    return min(dp[d][l], dp[d][r - (1 << d)]);
  }
  void init(const vt<T>& v) {
    dp = {v};
    for (int w = 2; w <= size(v); w += w) {
      dp.emplace_back(size(v) - w + 1);
      for (int i = 0; i + w <= size(v); i++)
        dp.back()[i] = query(i, i + w);
    }
  }
};
```

StaticRangeQuery.h

Description: Generic static range query for associative operations.
Usage: Define the desired operation
Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(1)$ query.

82b9ca, 34 lines

```
template<class T> struct RangeQuery {
  #define comb(a, b) (a) + (b)
  #define id 0
  int lg, n;
  vt<vt<T>> stor;
  vt<T> a;
  void fill(int l, int r, int ind) {
    if (ind < 0) return;
    int m = (l + r) / 2;
    T prod = id;
    FOR (i, m, r) stor[i][ind] = prod = comb(prod, a[i]);
    prod = id;
    ROF (i, l, m) stor[i][ind] = prod = comb(a[i], prod);
    fill(l, m, ind - 1);
    fill(m, r, ind - 1);
  }
  template <typename It>
  void build(It l, It r) {
    lg = 1;
    while ((1 << lg) < r - l) lg++;
    n = 1 << lg;
    a.resize(n, id);
    for (It i = l; i != r; i++) a[i - l] = *i;
    stor.resize(n, vt<T>(32 - __builtin_clz(n)));
    fill(0, n, lg - 1);
  }
  T query(int l, int r) {
    if (l == r) return a[l];
    int t = 31 - __builtin_clz(r ^ l);
    return comb(stor[l][t], stor[r][t]);
  }
  #undef id
  #undef comb
};
```

Matrix.h

Description: Basic operations on square matrices.
Usage: Matrix<int, 3> A;
A.d = {{{{1,2,3}}, {{4,5,6}}, {{7,8,9}}}};
array<int, 3> vec = {1,2,3};
vec = (A^N) * vec;

a85526, 26 lines

```
template<class T, int N> struct Matrix {
  typedef Matrix M;
  array<array<T, N>, N> d{};
  M operator*(const M& m) const {
    M a;
    FOR (i, N) FOR (j, N)
      FOR (k, N) a.d[i][k] += d[i][j] * m.d[j][k];
    return a;
  }
  array<T, N> operator*(const array<T, N>& vec) const {
    array<T, N> ret{};
    FOR (i, N) FOR (j, N) ret[i] += d[i][j] * vec[j];
    return ret;
  }
  M operator^(ll p) const {
    assert(p >= 0);
    M a, b(*this);
    FOR (i, N) a.d[i][i] = 1;
    while (p) {
      if (p & 1) a = a * b;
      b = b * b;
      p >>= 1;
    }
    return a;
  }
};
```

LineContainer.h

Description: Container where you can add lines of the form kx+m, and query maximum values at points x. Useful for dynamic programming ("convex hull trick").
Time: $\mathcal{O}(\log N)$

8ec1c7, 30 lines

```
struct Line {
  mutable ll k, m, p;
  bool operator<(const Line& o) const { return k < o.k; }
  bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
  // (for doubles, use inf = 1/.0, div(a,b) = a/b)
  static const ll inf = LLONG_MAX;
  ll div(ll a, ll b) { // floored division
    return a / b - ((a ^ b) < 0 && a % b); }
  bool isect(iterator x, iterator y) {
    if (y == end()) return x->p = inf, 0;
    if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
    else x->p = div(y->m - x->m, x->k - y->k);
    return x->p >= y->p;
  }
  void add(ll k, ll m) {
    auto z = insert({k, m, 0}), y = z++, x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
    while ((y = x) != begin() && (--x)->p >= y->p)
      isect(x, erase(y));
  }
  ll query(ll x) {
    assert(!empty());
    auto l = *lower_bound(x);
    return l.k * x + l.m;
  }
};
```

Trie.h

Description: maintains a set of ints with queries related to xor
Time: $\mathcal{O}(lg)$

98c78d, 72 lines

```
const int lg = 30;
using ptr = struct Node *;
struct Node {
  int cnt;
  ptr c[2];
};

int ins(ptr &n, int x, int i = lg) {
  if (!n) n = new Node {}; //  prevents dupes v
  return (i-- ? ins(n->c[1 & x >> i], x, i) : !n->cnt) ? ++n->cnt : 0;
}

void multi_ins(ptr &n, int x, int i = lg) {
  if (!n) n = new Node {};
  if (i--) multi_ins(n->c[1 & x >> i], x, i);
  n->cnt++;
}

int del(ptr n, int x, int i = lg) {
  return (n && (i-- ? del(n->c[1 & x >> i], x, i) : n->cnt)) ? n->cnt
        -- : 0;
}

int cnt(ptr n) { return n ? n->cnt : 0; }

// find y in n with minimum x ^ y
int qmin(ptr n, int x, int i = lg) {
  if (!i--) return 0;
  int b = 1 & x >> i;
  return cnt(n->c[b]) ? qmin(n->c[b], x, i) :
    qmin(n->c[!b], x, i) | (1 << i);
}

// find y in n with maximum x ^ y
int qmax(ptr n, int x, int i = lg) {
  if (!i--) return 0;
  int b = 1 & ~x >> i;
  return cnt(n->c[b]) ? qmax(n->c[b], x, i) | (1 << i) :
    qmax(n->c[!b], x, i);
}

// count y in n such that:
template<int sgn> // sgn = 0 for x ^ y < k, sgn = 1 for x ^ y > k
int count(ptr n, int x, int k, int i = lg) {
  if (!i-- || !cnt(n)) return 0;
  int b = 1 & (x ^ k) >> i;
  return ((1 & k >> i) ^ sgn ? cnt(n->c[!b]) : 0)
    + count<sgn>(n->c[b], x, k, i);
}

inline void push(ptr n, int i) { // lazy xor
  if (i && 1 & n->lazy >> (i - 1)) swap(n->c[0], n->c[1]);
  for (int j = 2; j--; n->c[j] && (n->c[j]->lazy ^= n->lazy));
  n->lazy = 0;
}

// you can directly add cnt if multiset - otherwise need to pull
int merge(ptr &l, ptr r, int i = lg) {
  if (!l) swap(l, r);
  if (!l) return 0;
  if (!r || i == 1) return l->cnt;
  push(l, i), push(r, i);
  l->cnt = 0;
  for (int j = 2; j--; l->cnt += merge(l->c[j], r->c[j], i - 1));
  return l->cnt;
```

```
}

int mex(ptr n, int i = lg) {
  if (!n || !i) return 0;
  push(n, i--);
  int b = n->c[0] && (n->c[0]->cnt == (1 << i));
  return mex(n->c[b], i) + (b << i);
}
```

Numerical (4)

4.1 Polynomials and recurrences

QuadRoots.h

Description: Calculates the roots of a quadratic equation with better precision. Returns the number of roots.
Time: $\mathcal{O}(1)$

cda769, 8 lines

```
int quadRoots(db a, db b, db c, pair<db, db> &out) {
  assert(a != 0);
  db disc = b * b - 4 * a * c;
  if (disc < 0) return 0;
  db sum = (b >= 0) ? -b - sqrt(disc) : -b + sqrt(disc);
  out = {sum / (2 * a), sum == 0 ? 0 : (2 * c) / sum};
  return 1 + (disc > 0);
}
```

Polynomial.h

e9feba, 17 lines

```
struct Poly {
  vt<db> a;
  db operator()(double x) const {
    double val = 0;
    for (int i = size(a); i--;) (val *= x) += a[i];
    return val;
  }
  void diff() {
    FOR (i, 1, size(a)) a[i - 1] = i * a[i];
    a.pop_back();
  }
  void divroot(double x0) {
    db b = a.back(), c; a.back() = 0;
    for (int i = size(a) - 1; i--;) c = a[i], a[i] = a[i + 1] * x0 + b
          , b = c;
    a.pop_back();
  }
};
```

PolyRoots.h

Description: Finds the real roots to a polynomial.
Usage: polyRoots({{2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0
Time: $\mathcal{O}\left(n^2 \log(1/\epsilon)\right)$

"Polynomial.h" f1a961, 23 lines

```
vt<db> poly_roots(Poly p, double xmin, double xmax) {
  if (size(p.a) == 2) { return {-p.a[0] / p.a[1]}; }
  vt<db> ret;
  Poly der = p;
  der.diff();
  auto dr = poly_roots(der, xmin, xmax);
  dr.push_back(xmin - 1);
  dr.push_back(xmax + 1);
  sort(all(dr));
  FOR (i, size(dr) - 1) {
    db l = dr[i], h = dr[i + 1];
    bool sign = p(l) > 0;
    if (sign ^ (p(h) > 0)) {
      FOR (it, 60) { // while (h - l > 1e-8)
        double m = (l + h) / 2, f = p(m);
        if ((f <= 0) ^ sign) l = m;
        else h = m;
      }
      ret.push_back((l + h) / 2);
    }
  }
  return ret;
}
```

PolyInterpolate.h

Description: Given n points (x[i], y[i]), computes an n-1-degree polynomial p that passes through them: $p(x) = a[0] * x^0 + ... + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi), k = 0 \dots n-1$.
Time: $\mathcal{O}\left(n^2\right)$

cc9fb9, 13 lines

```
using vd = vt<db>;
vd interpolate(vd x, vd y, int n) {
  vd res(n), temp(n);
  FOR (k, n - 1) FOR (i, k + 1, n)
    y[i] = (y[i] - y[k]) / (x[i] - x[k]);
  double last = 0; temp[0] = 1;
  FOR (k, n) FOR (i, n) {
    res[i] += y[k] * temp[i];
    swap(last, temp[i]);
    temp[i] -= last * x[k];
  }
  return res;
}
```

BerlekampMassey.h

Description: Recovers any n-order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\leq n$.
Usage: berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}
Time: $\mathcal{O}\left(N^2\right)$

"../number-theory/ModPow.h" ce4d42, 20 lines

```
vt<ll> berlekampMassey(vt<ll> s) {
  int n = size(s), L = 0, m = 0;
  vt<ll> C(n), B(n), T;
  C[0] = B[0] = 1;

  ll b = 1;
  FOR (i, n) { ++m;
    ll d = s[i] % mod;
    FOR (j, 1, L + 1) d = (d + C[j] * s[i - j]) % mod;
    if (!d) continue;
    T = C; ll coef = d * mpow(b, mod - 2) % mod;
    FOR (j, m, n) C[j] = (C[j] - coef * B[j - m]) % mod;
    if (2 * L > i) continue;
    L = i + 1 - L; B = T; b = d; m = 0;
  }

  C.resize(L + 1); C.erase(C.begin());
  for (ll &x : C) x = (mod - x) % mod;
  return C;
}
```

LinearRecurrence.h

Description: Generates the k'th term of an n-order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \dots \geq n-1]$ and $tr[0 \dots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp–Massey.
Usage: linearRec({0, 1}, {1, 1}, k) // k'th Fibonacci number
Time: $\mathcal{O}\left(n^2 \log k\right)$

602ede, 26 lines

```
using Poly = vt<ll>;
ll linearRec(Poly S, Poly tr, ll k) {
  int n = size(tr);

  auto combine = [&](Poly a, Poly b) {
    Poly res(n * 2 + 1);
    FOR (i, n + 1) FOR (j, n + 1)
      res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
    for (int i = 2 * n; i > n; --i) FOR (j, n)
      res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
    res.resize(n + 1);
    return res;
  };

  Poly pol(n + 1), e(pol);
  pol[0] = e[1] = 1;

  for (++k; k; k /= 2) {
    if (k % 2) pol = combine(pol, e);
    e = combine(e, e);
  }

  ll res = 0;
  FOR (i, n) res = (res + pol[i + 1] * S[i]) % mod;
  return res;
}
```

4.2 Optimization

GoldenSectionSearch.h

Description: Finds the argument minimizing the function f in the interval $[a, b]$ assuming f is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is eps. Works equally well for maximization with a small change in the code. See TernarySearch.h in the Various chapter for a discrete version.
Usage: db func(db x) { return 4+x+.3*x*x; }
db xmin = gss(-1000,1000,func);
Time: $\mathcal{O}(\log((b-a)/\epsilon))$

dad647, 14 lines

```
db gss(db a, db b, db (*f)(db)) {
  db r = (sqrt(5)-1)/2, eps = 1e-7;
  db x1 = b - r*(b-a), x2 = a + r*(b-a);
  db f1 = f(x1), f2 = f(x2);
  while (b - a > eps)
    if (f1 < f2) { //change to > to find maximum
      b = x2; x2 = x1; f2 = f1;
      x1 = b - r*(b-a); f1 = f(x1);
    } else {
      a = x1; x1 = x2; f1 = f2;
      x2 = a + r*(b-a); f2 = f(x2);
    }
  return a;
}
```

Integrate.h

Description: Simple integration of a function over an interval using Simpson's rule. The error should be proportional to h^4, although in practice you will want to verify that the result is stable to desired precision when epsilon changes.

4756fc, 7 lines

```
template<class F>
double quad(double a, double b, F f, const int n = 1000) {
  double h = (b - a) / 2 / n, v = f(a) + f(b);
  rep(i,1,n*2)
    v += f(a + i*h) * (i&1 ? 4 : 2);
  return v * h / 3;
}
```

IntegrateAdaptive.h

Description: Fast integration using an adaptive Simpson's rule.

Usage: double sphereVolume = quad(-1, 1, [](double x) {
return quad(-1, 1, [&](double y) {
return quad(-1, 1, [&](double z) {
return x*x + y*y + z*z < 1; });});});

92dd79, 15 lines

```
typedef double d;
#define S(a,b) (f(a) + 4*f((a+b) / 2) + f(b)) * (b-a) / 6

template <class F>
d rec(F& f, d a, d b, d eps, d S) {
  d c = (a + b) / 2;
  d S1 = S(a, c), S2 = S(c, b), T = S1 + S2;
  if (abs(T - S) <= 15 * eps || b - a < 1e-10)
    return T + (T - S) / 15;
  return rec(f, a, c, eps / 2, S1) + rec(f, c, b, eps / 2, S2);
}
template<class F>
d quad(d a, d b, F f, d eps = 1e-8) {
  return rec(f, a, b, eps, S(a, b));
}
```

Simplex.h

Description: Solves a general linear maximization problem: maximize $c^T x$ subject to $Ax \le b$, $x \ge 0$. Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of $c^T x$ otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that $x = 0$ is viable.
Usage: vvd A = {{1,-1}, {-1,1}, {-1,-2}};
vd b = {1,1,-4}, c = {-1,-1}, x;
T val = LPSolver(A, b, c).solve(x);
Time: $\mathcal{O}(NM * \#pivots)$, where a pivot may be e.g. an edge relaxation. $\mathcal{O}(2^n)$ in the general case.

4ae05c, 60 lines

```
#define mp make_pair
using T = db;
using vd = vt<db>;
using vvd = vt<vd>;
const db eps = 1e-9;
const int inf = 1e9;

#define ltj(X) if (s == -1 || mp(X[j], N[j]) < mp(X[s], N[s])) s = j
struct LPSolver {
  int m, n;
  vt<int> N, B;
  vvd D;
  LPSolver(const vvd& A, const vd& b, const vd& c) :
    m(size(b)), n(size(c)), N(n + 1), B(m), D(m + 2, vd(n + 2)) {
    F0R(i, m) F0R(j, n) D[i][j] = A[i][j];
    F0R(i, m) B[i] = n+i, D[i][n] = -1, D[i][n + 1] = b[i];
    F0R(j, n) N[j] = j, D[m][j] = -c[j];
    N[n] = -1; D[m + 1][n] = 1;
  }
  void pivot(int r, int s) {
    T inv = 1 / D[r][s];
    F0R(i, m + 2) if (i != r && abs(D[i][s]) > eps) {
      T binv = D[i][s]*inv;
      F0R (j, n + 2) if (j != s) D[i][j] -= D[r][j]*binv;
      D[i][s] = -binv;
    }
    D[r][s] = 1; F0R(j, n + 2) D[r][j] *= inv; // scale r-th row
    swap(B[r],N[s]);
  }
  bool simplex(int phase) {
    int x = m + phase - 1;
    while (1) {
      int s = -1; F0R (j, n + 1) if (N[j] != -phase) ltj(D[x]);
      if (D[x][s] >= -eps) return 1;
      int r = -1;
      F0R (i, m) {
        if (D[i][s] <= eps) continue;
        if (r == -1 || mp(D[i][n + 1] / D[i][s], B[i])
              < mp(D[r][n + 1] / D[r][s], B[r])) r = i;
      }
      if (r == -1) return 0;
      pivot(r, s);
    }
  }
  T solve(vd &x) {
    int r = 0; FOR (i, 1, m) if (D[i][n + 1] < D[r][n + 1]) r = i;
    if (D[r][n + 1] < -eps) {
      pivot(r,n);
      assert(simplex(2));
      if (D[m + 1][n + 1] < -eps) return -inf;
      F0R (i, m) if (B[i] == -1) {
        int s = 0; FOR (j, 1, n + 1) ltj(D[i]);
        pivot(i, s);
      }
    }
    bool ok = simplex(1); x = vd(n);
    F0R (i, m) if (B[i] < n) x[B[i]] = D[i][n + 1];
    return ok ? D[m][n + 1] : inf;
  }
};
```

4.3 Matrices

Determinant.h

Description: Calculates determinant of a matrix. Destroys the matrix.
Time: $\mathcal{O}(N^3)$

ef2d92, 15 lines

```
db det(vt<vt<db>>& a) {
  int n = size(a); db res = 1;
  F0R (i, n) {
    int b = i;
    FOR (j, i + 1, n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
    if (i != b) swap(a[i], a[b]), res *= -1;
    res *= a[i][i];
    if (res == 0) return 0;
    FOR (j, i + 1, n) {
      double v = a[j][i] / a[i][i];
      if (v != 0) FOR (k, i + 1, n) a[j][k] -= v * a[i][k];
    }
  }
  return res;
}
```

IntDeterminant.h

Description: Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.
Time: $\mathcal{O}(N^3)$

3313dc, 18 lines

```
const ll mod = 12345;
ll det(vector<vector<ll>>& a) {
  int n = sz(a); ll ans = 1;
  rep(i,0,n) {
    rep(j,i+1,n) {
      while (a[j][i] != 0) { // gcd step
        ll t = a[i][i] / a[j][i];
        if (t) rep(k,i,n)
          a[i][k] = (a[i][k] - a[j][k] * t) % mod;
        swap(a[i], a[j]);
        ans *= -1;
      }
    }
    ans = ans * a[i][i] % mod;
    if (!ans) return 0;
  }
  return (ans + mod) % mod;
}
```

SolveLinear.h

Description: Solves $A * x = b$. If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in A and b is lost.
Time: $\mathcal{O}(n^2 m)$

0ff65f, 38 lines

```
using vd = vt<db>;
const double eps = 1e-12;

int solveLinear(vt<vd>& A, vd& b, vd& x) {
  int n = size(A), m = size(x), rank = 0, br, bc;
  if (n) assert(size(A[0]) == m);
  vi col(m); iota(all(col), 0);

  F0R (i, n) {
    double v, bv = 0;
    FOR (r, i, n) FOR (c, i, m)
      if ((v = fabs(A[r][c])) > bv)
        br = r, bc = c, bv = v;
    if (bv <= eps) {
      FOR (j, i, n) if (fabs(b[j]) > eps) return -1;
      break;
    }
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    F0R (j, n) swap(A[j][i], A[j][bc]);
    bv = 1 / A[i][i];
    FOR (j, i + 1, n) {
      double fac = A[j][i] * bv;
      b[j] -= fac * b[i];
      FOR (k, i + 1, m) A[j][k] -= fac * A[i][k];
    }
    rank++;
  }

  x.assign(m, 0);
  for (int i = rank; i--;) {
    b[i] /= A[i][i];
    x[col[i]] = b[i];
    F0R (j, i) b[j] -= A[j][i] * b[i];
  }
  return rank; // (multiple solutions if rank < m)
}
```

SolveLinear2.h

Description: To get all uniquely determined values of x back from SolveLinear, make the following changes:

"SolveLinear.h" 77b9bb, 7 lines

```
F0R (j, n) if (j != i) // instead of FOR (j, i + 1, n)
// ... then at the end:
x.assign(m, undefined);
F0R (i, rank) {
  FOR (j, rank, m) if (fabs(A[i][j]) > eps) goto fail;
  x[col[i]] = b[i] / A[i][i];
fail:; }
```

SolveLinearBinary.h

Description: Solves $Ax = b$ over $\mathbb{F}_2$. If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys A and b.
Time: $\mathcal{O}(n^2 m)$

042fea, 34 lines

```
typedef bitset<1000> bs;

int solveLinear(vt<bs>& A, vi& b, bs& x, int m) {
  int n = size(A), rank = 0, br;
  assert(m <= size(x));
  vi col(m); iota(all(col), 0);
```

```
  FOR (i, n) {
    FOR (br, i, n) if (A[br].any()) break;
    if (br == n) {
      FOR (j, i, n) if(b[j]) return -1;
      break;
    }
    int bc = (int) A[br]._Find_next(i - 1);
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    FOR (j, n) if (A[j][i] != A[j][bc]) {
      A[j].flip(i); A[j].flip(bc);
    }
    FOR (j, i + 1, n) if (A[j][i]) {
      b[j] ^= b[i];
      A[j] ^= A[i];
    }
    rank++;
  }

  x = bs();
  for (int i = rank; i--;) {
    if (!b[i]) continue;
    x[col[i]] = 1;
    FOR (j, i) b[j] ^= A[j][i];
  }
  return rank; // (multiple solutions if rank < m)
}
```

MatrixInverse.h

Description: Invert matrix A. Returns rank; result is stored in A unless singular (rank < n). Can easily be extended to prime moduli; for prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$ where A^{-1} starts as the inverse of A mod p, and k is doubled in each step.
Time: $\mathcal{O}(n^3)$

cb738d, 35 lines

```
int matInv(vt<vt<db>>& A) {
  int n = size(A); vi col(n);
  vt<vt<db>> tmp(n, vt<db>(n));
  FOR (i, n) tmp[i][i] = 1, col[i] = i;

  FOR (i, n) {
    int r = i, c = i;
    FOR (j, i, n) FOR (k, i, n)
      if (fabs(A[j][k]) > fabs(A[r][c]))
        r = j, c = k;
    if (fabs(A[r][c]) < 1e-12) return i;
    A[i].swap(A[r]); tmp[i].swap(tmp[r]);
    FOR (j, n)
      swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
    swap(col[i], col[c]);
    double v = A[i][i];
    FOR (j, i + 1, n) {
      double f = A[j][i] / v;
      A[j][i] = 0;
      FOR (k, i + 1, n) A[j][k] -= f * A[i][k];
      FOR (k, 0, n) tmp[j][k] -= f * tmp[i][k];
    }
    FOR (j, i + 1, n) A[i][j] /= v;
    FOR (j, n) tmp[i][j] /= v;
    A[i][i] = 1;
  }

  for (int i = n - 1; i > 0; --i) FOR (j, i) {
    double v = A[j][i];
    FOR (k, n) tmp[j][k] -= v * tmp[i][k];
  }

  FOR (i, n) FOR (j, n) A[col[i]][col[j]] = tmp[i][j];
  return n;
}
```

MatrixInverse-mod.h

Description: Invert matrix A modulo a prime. Returns rank; result is stored in A unless singular (rank < n). For prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$ where A^{-1} starts as the inverse of A mod p, and k is doubled in each step.
Time: $\mathcal{O}(n^3)$

"../number-theory/ModPow.h" 0b7b13, 37 lines

```
int matInv(vector<vector<ll>>& A) {
  int n = sz(A); vi col(n);
  vector<vector<ll>> tmp(n, vector<ll>(n));
  rep(i,0,n) tmp[i][i] = 1, col[i] = i;

  rep(i,0,n) {
    int r = i, c = i;
    rep(j,i,n) rep(k,i,n) if (A[j][k]) {
      r = j; c = k; goto found;
    }
    return i;
found:
    A[i].swap(A[r]); tmp[i].swap(tmp[r]);
    rep(j,0,n)
      swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
    swap(col[i], col[c]);
    ll v = modpow(A[i][i], mod - 2);
    rep(j,i+1,n) {
      ll f = A[j][i] * v % mod;
      A[j][i] = 0;
      rep(k,i+1,n) A[j][k] = (A[j][k] - f*A[i][k]) % mod;
      rep(k,0,n) tmp[j][k] = (tmp[j][k] - f*tmp[i][k]) % mod;
    }
    rep(j,i+1,n) A[i][j] = A[i][j] * v % mod;
    rep(j,0,n) tmp[i][j] = tmp[i][j] * v % mod;
    A[i][i] = 1;
  }

  for (int i = n-1; i > 0; --i) rep(j,0,i) {
    ll v = A[j][i];
    rep(k,0,n) tmp[j][k] = (tmp[j][k] - v*tmp[i][k]) % mod;
  }

  rep(i,0,n) rep(j,0,n)
    A[col[i]][col[j]] = tmp[i][j] % mod + (tmp[i][j] < 0)*mod;
  return n;
}
```

4.4 Convolutions

FastFourierTransform.h

Description: fft(a) computes $\hat{f}(k) = \sum_x a[x]\exp(2\pi i \cdot kx/N)$ for all k. N must be a power of 2. Useful for convolution: `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by n, reverse(start+1, end), FFT back. Rounding is safe if $(\sum a_i^2 + \sum b_i^2)\log_2 N < 9 \cdot 10^{14}$ (in practice 10^{16}; higher for random inputs). Otherwise, use NTT/FFTMod.
Time: $\mathcal{O}(N \log N)$ with $N = |A| + |B|$ ($\sim 0.2s$ for $N = 2^{20}$)

636a47, 36 lines

```
using C = complex<db>;
using vd = vt<db>;

void fft(vt<C> &a) {
  int n = size(a), L = 31 - __builtin_clz(n);
  static vt<complex<long double>> R(2, 1);
  static vt<C> rt(2, 1);  // (^ 10% faster if db)
  for (static int k = 2; k < n; k *= 2) {
    R.resize(n); rt.resize(n);
    auto x = polar(1.0L, acos(-1.0L) / k);
    FOR (i, k, 2 * k) rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
  }
  vi rev(n);
  FOR (i, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
  FOR (i, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
  for (int k = 1; k < n; k *= 2)
    for (int i = 0; i < n; i += 2 * k) FOR (j, k) {
      C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-rolled)
      a[i + j + k] = a[i + j] - z;
      a[i + j] += z;
    }
}
vd conv(const vd &a, const vd &b) {
  if (a.empty() || b.empty()) return {};
  vd res(size(a) + size(b) - 1);
  int L = 32 - __builtin_clz(size(res)), n = 1 << L;
  vt<C> in(n), out(n);
  copy(all(a), begin(in));
  FOR (i, size(b)) in[i].imag(b[i]);
  fft(in);
  for (C& x : in) x *= x;
  FOR (i, n) out[i] = in[-i & (n - 1)] - conj(in[i]);
  fft(out);
  FOR (i, size(res)) res[i] = imag(out[i]) / (4 * n);
  return res;
}
```

FastFourierTransformMod.h

Description: Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$ (in practice 10^{16} or higher). Inputs must be in $[0, \text{mod})$.
Time: $\mathcal{O}(N \log N)$, where $N = |A| + |B|$ (twice as slow as NTT or FFT) (but seemed +10% on yosupo?)

"FastFourierTransform.h" c4a07e, 22 lines

```
using vl = vt<ll>;
template<int M> vl convMod(const vl &a, const vl &b) {
  if (a.empty() || b.empty()) return {};
  vl res(size(a) + size(b) - 1);
  int B = 32 - __builtin_clz(size(res)), n = 1 << B, cut = int(sqrt(M)
      );
  vt<C> L(n), R(n), outs(n), outl(n);
  FOR (i, size(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
  FOR (i, size(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
  fft(L), fft(R);
  FOR (i, n) {
    int j = -i & (n - 1);
    outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
    outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
  }
  fft(outl), fft(outs);
  FOR (i, size(res)) {
    ll av = ll(real(outl[i]) + .5), cv = ll(imag(outs[i]) + .5);
    ll bv = ll(imag(outl[i]) + .5) + ll(real(outs[i]) + .5);
    res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
  }
  return res;
}
```

NumberTheoreticTransform.h

Description: ntt(a) computes $\hat{f}(k) = \sum_x a[x]g^{xk}$ for all k, where $g = \text{root}^{(mod-1)/N}$. N must be a power of 2. Useful for convolution modulo specific nice primes of the form $2^a b + 1$, where the convolution result has size at most 2^a. For arbitrary modulo, see FFTMod. `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For manual convolution: NTT the inputs, multiply pointwise, divide by n, reverse(start+1, end), NTT back. Inputs must be in [0, mod).
Time: $\mathcal{O}(N \log N)$ with $N = |A| + |B|$ ($\sim 0.2s$ for $N = 2^{20}$)

"../number-theory/ModPow.h" 53740f, 37 lines

```
const ll root = 62; // = 998244353
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 << 21
// and 483 << 21 (same root). The last two are > 10^9.

template<class T>
void ntt(vt<T> &a) {
  int n = size(a), L = 31 - __builtin_clz(n);
  static vt<ll> rt(2, 1);
  for (static int k = 2, s = 2; k < n; k *= 2, s++) {
    rt.resize(n);
    ll z[] = {1, mpow(root, mod >> s)};
    FOR (i, k, 2 * k) rt[i] = rt[i / 2] * z[i & 1] % mod;
  }
  vi rev(n);
  FOR (i, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
  FOR (i, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
  for (int k = 1; k < n; k *= 2)
    for (int i = 0; i < n; i += 2 * k) FOR (j, 0, k) {
      T z = (ll) rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
      a[i + j + k] = ai - z + (z > ai ? mod : 0);
      ai += (ai + z >= mod ? z - mod : z);
    }
}

template<class T>
vt<T> conv(const vt<T> &a, const vt<T> &b) {
  if (a.empty() || b.empty()) return {};
  int s = size(a) + size(b) - 1, B = 32 - __builtin_clz(s),
    n = 1 << B;
  int inv = mpow(n, mod - 2);
  vt<T> L(a), R(b), out(n);
  L.resize(n), R.resize(n);
  ntt(L), ntt(R);
  FOR (i, n) out[-i & (n - 1)] = (ll) L[i] * R[i] % mod * inv % mod;
  ntt(out);
  return {out.begin(), out.begin() + s};
}
```

FastSubsetTransform.h

Description: Transform to a basis with fast convolutions of the form $c[z] = \sum_{z=x\oplus y} a[x] \cdot b[y]$, where $\oplus$ is one of AND, OR, XOR. The size of a must be a power of two. Replace with long longs and do operations under mod if needed.
Time: $\mathcal{O}(N \log N)$

97380b, 18 lines

```
// don't forget MOD!
using pi = pair<int, int>;
void FST(vi &a, bool inv) {
  for (int n = size(a), step = 1; step < n; step *= 2) {
    for (int i = 0; i < n; i += 2 * step) FOR (j, i, i + step) {
      int &u = a[j], &v = a[j + step]; tie(u, v) =
        inv ? pi(v - u, u) : pi(v, u + v); // AND
        inv ? pi(v, u - v) : pi(u + v, u); // OR
        pi(u + v, u - v);          // XOR
    }
  }
  if (inv) for (int& x : a) x /= size(a); // XOR only
}
vi conv(vi a, vi b) {
  FST(a, 0); FST(b, 0);
  FOR (i, size(a)) a[i] *= b[i];
  FST(a, 1); return a;
}
```

MinPlusConvolution.h

Description: funny thing
Time: $\mathcal{O}(N)$

e9c1a1, 22 lines

```
// res[k] = min{0 <= i <= k}(arr1[i] + arr2[k - i])
// arr1, arr2 must both be convex arrays!
// O(n)
template<typename T>
vector<T> min_plus_convolution_convex_convex(const vector<T>& arr1,
    const vector<T>& arr2) {
  if (arr1.empty() || arr2.empty()) return arr1.empty() ? arr2 : arr1;
  const size_t n1 = arr1.size();
  const size_t n2 = arr2.size();
  for (size_t i = 1; i + 1 < n1; ++i) assert(arr1[i] - arr1[i - 1] <=
      arr1[i + 1] - arr1[i] && "arr1 must be convex!");
  for (size_t i = 1; i + 1 < n2; ++i) assert(arr2[i] - arr2[i - 1] <=
      arr2[i + 1] - arr2[i] && "arr2 must be convex!");
  vector<T> res(n1 + n2 - 1);
  size_t i1 = 0, i2 = 0;
  res[0] = arr1[0] + arr2[0];
  for (size_t i = 1; i < n1 + n2 - 1; ++i) {
    if (i2 == n2 - 1 || (i1 != n1 - 1 && arr1[i1 + 1] + arr2[i2] <
        arr1[i1] + arr2[i2 + 1])) {
      res[i] = arr1[++i1] + arr2[i2];
    } else {
      res[i] = arr1[i1] + arr2[++i2];
    }
  }
  return res;
}
```

Number theory (5)

5.1 Modular arithmetic

ModularArithmetic.h

Description: Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

"euclid.h" 35bfea, 18 lines

```
const ll mod = 17; // change to something else
struct Mod {
  ll x;
  Mod(ll xx) : x(xx) {}
  Mod operator+(Mod b) { return Mod((x + b.x) % mod); }
  Mod operator-(Mod b) { return Mod((x - b.x + mod) % mod); }
  Mod operator*(Mod b) { return Mod((x * b.x) % mod); }
  Mod operator/(Mod b) { return *this * invert(b); }
  Mod invert(Mod a) {
    ll x, y, g = euclid(a.x, mod, x, y);
    assert(g == 1); return Mod((x + mod) % mod);
  }
  Mod operator^(ll e) {
    if (!e) return Mod(1);
    Mod r = *this ^ (e / 2); r = r * r;
    return e&1 ? *this * r : r;
  }
};
```

ModInverse.h

Description: Pre-computation of modular inverses. Assumes LIM $\leq$ mod and that mod is a prime.

e403d4, 3 lines

```
const ll mod = 1000000007, LIM = 200000;
ll* inv = new ll[LIM] - 1; inv[1] = 1;
FOR (i, 2, LIM) inv[i] = mod - (mod / i) * inv[mod % i] % mod;
```

ModPow.h

d63292, 8 lines

```
const ll mod = 1000000007; // faster if const

ll mpow(ll b, ll e) {
  ll ans = 1;
  for (; e; b = b * b % mod, e /= 2)
    if (e & 1) ans = ans * b % mod;
  return ans;
}
```

ModLog.h

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$, or -1 if no such x exists. modLog(a,1,m) can be used to calculate the order of a.
Time: $\mathcal{O}(\sqrt{m})$

c040b8, 11 lines

```
ll modLog(ll a, ll b, ll m) {
  ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
  unordered_map<ll, ll> A;
  while (j <= n && (e = f = e * a % m) != b % m)
    A[e * b % m] = j++;
  if (e == b % m) return j;
  if (__gcd(m, e) == __gcd(m, b))
    rep(i,2,n+2) if (A.count(e = e * f % m))
      return n * i - A[e];
  return -1;
}
```

ModSum.h

Description: Sums of mod'ed arithmetic progressions.
modsum(to, c, k, m) $= \sum_{i=0}^{\mathrm{to}-1} (ki + c)\%m$. divsum is similar but for floored division.
Time: $\log(m)$, with a large constant.

5c5bc5, 16 lines

```
typedef unsigned long long ull;
ull sumsq(ull to) { return to / 2 * ((to-1) | 1); }

ull divsum(ull to, ull c, ull k, ull m) {
  ull res = k / m * sumsq(to) + c / m * to;
  k %= m; c %= m;
  if (!k) return res;
  ull to2 = (to * k + c) / m;
  return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k);
}

ll modsum(ull to, ll c, ll k, ll m) {
  c = ((c % m) + m) % m;
  k = ((k % m) + m) % m;
  return to * c + k * sumsq(to) - m * divsum(to, c, k, m);
}
```

ModMulLL.h

Description: Calculate $a \cdot b \bmod c$ (or $a^b \bmod c$) for $0 \leq a, b \leq c \leq 7.2 \cdot 10^{18}$.
Time: $\mathcal{O}(1)$ for modmul, $\mathcal{O}(\log b)$ for modpow

bb4d60, 11 lines

```
typedef unsigned long long ull;
ull mmul(ull a, ull b, ull M) {
  ll ret = a * b - M * ull(1.L / M * a * b);
  return ret + M * (ret < 0) - M * (ret >= (ll) M);
}
ull mpow(ull b, ull e, ull mod) {
  ull ans = 1;
  for (; e; b = mmul(b, b, mod), e /= 2)
    if (e & 1) ans = mmul(ans, b, mod);
  return ans;
}
```

ModSqrt.h

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).
Time: $\mathcal{O}(\log^2 p)$ worst case, $\mathcal{O}(\log p)$ for most p

"ModPow.h" 4ba904, 24 lines

```
ll sqrt(ll a, ll p) {
  a %= p; if (a < 0) a += p;
  if (a == 0) return 0;
```

```
  assert(mpow(a, (p-1)/2, p) == 1); // else no solution
  if (p % 4 == 3) return mpow(a, (p+1)/4, p);
  // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
  ll s = p - 1, n = 2;
  int r = 0, m;
  while (s % 2 == 0)
    ++r, s /= 2;
  while (mpow(n, (p - 1) / 2, p) != p - 1) ++n;
  ll x = mpow(a, (s + 1) / 2, p);
  ll b = mpow(a, s, p), g = mpow(n, s, p);
  for (;; r = m) {
    ll t = b;
    for (m = 0; m < r && t != 1; ++m)
      t = t * t % p;
    if (m == 0) return x;
    ll gs = mpow(g, 1LL << (r - m - 1), p);
    g = gs * gs % p;
    x = x * gs % p;
    b = b * g % p;
  }
}
```

5.2 Primality

Sieve.h

Description: Linear sieve implementation. Around 0.5s for 1e8. Also computes smallest prime divisor for each number in lp.

6f6405, 22 lines

```
const int mx = 1e8; // ~0.5s
vi lp(mx + 1), primes;

FOR (i, 2, mx + 1) {
  if (lp[i] == 0) lp[i] = i, primes.pb(i);
  for (int j = 0; i * primes[j] <= mx; j++) {
    lp[i * primes[j]] = primes[j];
    // store i to avoid division when factoring
    if (primes[j] == lp[i]) break;
  }
}

auto factor = [&] (int x) {
  vt<pair<int, int>> res;
  while (x > 1) {
    if (!size(res) || res.back().first != lp[x])
      res.pb({lp[x], 0});
    res.back().second++;
    x /= lp[x]; // can avoid division by storing extra array
  }
  return res;
};
```

FactorSqrt.h

Description: Easy factorisation. Returns pairs of prime factor, multiplicity

Time: $\mathcal{O}\left(\sqrt{N}\right)$.

56249c, 15 lines

```
vt<pair<int, int>> factor(int x) {
  if (x == 1) return {};
  vt<pair<int, int>> res;
  for (int i = 2; i * i <= x; i++) {
    if (x % i == 0) {
      res.pb({i, 0});
      while (x % i == 0) {
        res.back().second++;
        x /= i;
      }
    }
  }
  if (x > 1) res.pb({x, 1});
  return res;
}
```

FastEratosthenes.h

Description: Prime sieve for generating all primes smaller than LIM.

Time: LIM=1e9 $\approx$ 1.5s

6b2912, 20 lines

```
const int LIM = 1e6;
bitset<LIM> isPrime;
vi eratosthenes() {
  const int S = (int)round(sqrt(LIM)), R = LIM / 2;
  vi pr = {2}, sieve(S+1); pr.reserve(int(LIM/log(LIM)*1.1));
  vector<pii> cp;
  for (int i = 3; i <= S; i += 2) if (!sieve[i]) {
    cp.push_back({i, i * i / 2});
    for (int j = i * i; j <= S; j += 2 * i) sieve[j] = 1;
  }
  for (int L = 1; L <= R; L += S) {
    array<bool, S> block{};
    for (auto &[p, idx] : cp)
      for (int i=idx; i < S+L; idx = (i+=p)) block[i-L] = 1;
    rep(i,0,min(S, R - L))
      if (!block[i]) pr.push_back((L + i) * 2 + 1);
  }
  for (int i : pr) isPrime[i] = 1;
  return pr;
}
```

MillerRabin.h

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.

Time: 7 times the complexity of $a^b \mod c$.

"ModMulLL.h" 168797, 12 lines

```
bool isPrime(ull n) {
  if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
  ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},
    s = __builtin_ctzll(n - 1), d = n >> s;
  for (ull a : A) {   // ^ count trailing zeroes
    ull p = mpow(a % n, d, n), i = s;
    while (p != 1 && p != n - 1 && a % n && i--)
      p = mmul(p, p, n);
    if (p != n-1 && i != s) return 0;
  }
  return 1;
}
```

Factor.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}). Consider manually trying small primes for better runtime. In one second: average 350k numbers at 1e9, 200k numbers at 1e12, and 100k at 1e16.

Time: $\mathcal{O}\left(n^{1/4}\right)$, less for numbers with small factors.

"ModMulLL.h", "MillerRabin.h" bf9dbb, 18 lines

```
ull pollard(ull n) {
  ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;
  auto f = [&](ull x) { return mmul(x, x, n) + i; };
  while (t++ % 40 || __gcd(prd, n) == 1) {
    if (x == y) x = ++i, y = f(x);
    if ((q = mmul(prd, max(x,y) - min(x,y), n))) prd = q;
    x = f(x), y = f(f(y));
  }
  return __gcd(prd, n);
}
vector<ull> factor(ull n) {
  if (n == 1) return {};
  if (isPrime(n)) return {n};
  ull x = pollard(n);
  auto l = factor(x), r = factor(n / x);
  l.insert(l.end(), all(r));
  return l;
}
```

PrimeCnt.h

Description: Counts number of primes up to N. Can also count sum of primes.

Time: $\mathcal{O}\left(N^{3/4}/\log N\right)$, 60ms for $N = 10^{11}$, 2.5s for $N = 10^{13}$

c04e96, 20 lines

```
ll count_primes(ll N) { // count_primes(1e13) == 346065536839
  if (N <= 1) return 0;
  int sq = (int)sqrt(N);
  vl big_ans((sq+1)/2), small_ans(sq+1);
  FOR(i,1,sq+1) small_ans[i] = (i-1)/2;
  FOR(i,sz(big_ans)) big_ans[i] = (N/(2*i+1)-1)/2;
  vb skip(sq+1); int prime_cnt = 0;
  for (int p = 3; p <= sq; p += 2) if (!skip[p]) { // primes
    for (int j = p; j <= sq; j += 2*p) skip[j] = 1;
    FOR(j,min((ll)sz(big_ans),(N/p/p+1)/2)) {
      ll prod = (ll)(2*j+1)*p;
      big_ans[j] -= (prod > sq ? small_ans[(double)N/prod]
              : big_ans[prod/2])-prime_cnt;
    }
    for (int j = sq, q = sq/p; q >= p; --q) for (;j >= q*p;--j)
      small_ans[j] -= small_ans[q]-prime_cnt;
    ++prime_cnt;
  }
  return big_ans[0]+1;
}
```

5.3 Divisibility

euclid.h

Description: Finds two integers x and y, such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in __gcd instead. If a and b are coprime, then x is the inverse of $a \pmod b$.

33ba8f, 5 lines

```
ll euclid(ll a, ll b, ll &x, ll &y) {
  if (!b) return x = 1, y = 0, a;
  ll d = euclid(b, a % b, y, x);
  return y -= a/b * x, d;
}
```

CRT.h

Description: Chinese Remainder Theorem.

crt(a, m, b, n) computes x such that $x \equiv a \pmod m$, $x \equiv b \pmod n$. If $|a| < m$ and $|b| < n$, x will obey $0 \le x < \mathrm{lcm}(m, n)$. Assumes $mn < 2^{62}$.

Time: $\log(n)$

"euclid.h" 04d93a, 7 lines

```
ll crt(ll a, ll m, ll b, ll n) {
  if (n > m) swap(a, b), swap(m, n);
  ll x, y, g = euclid(m, n, x, y);
  assert((a - b) % g == 0); // else no solution
  x = (b - a) % n * x % n / g * m + a;
  return x < 0 ? x + m*n/g : x;
}
```

5.3.1 Bézout's identity

For $a \neq$, $b \neq 0$, then $d = gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

phiFunction.h

Description: *Euler's* ϕ function is defined as $\phi(n) := \#$ of positive integers $\le n$ that are coprime with n. $\phi(1) = 1$, p prime $\Rightarrow \phi(p^k) = (p-1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1} p_2^{k_2} ... p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1} ... (p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n}(1 - 1/p)$.
$\sum_{d|n} \phi(d) = n$, $\sum_{1 \le k \le n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$
Euler's thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}$.
Fermat's little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod{p}$ $\forall a$.

5bf43c, 8 lines

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi() {
  F0R (i, LIM) phi[i] = i & 1 ? i : i / 2;
  for (int i = 3; i < LIM; i += 2) if(phi[i] == i)
    for (int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}
```

5.4 Fractions

5.4.1 Continued Fractions

Let $a_0, a_1, \ldots, a_n \in \mathbb{Z}$ and $a_1, a_2, \ldots, a_n \ge 1$. Then the expression

$$r = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}},$$

is called the *continued fraction representation* of the rational number r and is denoted shortly as

$$r = [a_0; a_1, a_2, \ldots, a_k].$$

If $a_0 > 0$ then $\frac{1}{r}$ is given by sticking a 0 in front. If $a_0 = 0$ then ditch the first term.
$[a_0, a_1, \ldots, a_n - 1]$ gives the run-length encoding (right first) of r's path in the Stern-Brocot tree.
$r_k = [a_0; a_1, a_2, \ldots, a_k] = \frac{p_k}{q_k}$ is called the k-th *convergent* of r.
Semiconvergents are convergents but you can lower the last number down to 1.
Consider the convex hull of points above and below $y = rx$. Odd convergents $(q_k; p_k)$ give vertices of the upper hull, while even convergents give the vertices of the lower hull.
The set of semiconvergents = set of lattice points on the hulls.
Combine these facts with Pick's theorem for funny stuff.

SBT.h

Description: Given p/q, find the shorter of its two unique continued fraction representations. The other representation is given by `vec.back()-, vec.pb(1);`
Time: $\mathcal{O}(\log N)$

fb1c2d, 8 lines

```
vi cont_frac(int p, int q) {
  vi a;
  while (q) {
    a.push_back(p / q);
    tie(p, q) = make_pair(q, p % q);
  }
  return a;
}
```

ContinuedFractions.h

Description: Given N and a real number $x \ge 0$, finds the closest rational approximation p/q with $p, q \le N$. It will obey $|p/q - x| \le 1/qN$.
For consecutive convergents, $p_{k+1}q_k - q_{k+1}p_k = (-1)^k$. ($p_k/q_k$ alternates between $> x$ and $< x$.) If x is rational, y eventually becomes ∞; if x is the root of a degree 2 polynomial the a's eventually become cyclic.
Time: $\mathcal{O}(\log N)$

dd6c5e, 21 lines

```
typedef double d; // for N ~ 1e7; long double for N ~ 1e9
pair<ll, ll> approximate(d x, ll N) {
  ll LP = 0, LQ = 1, P = 1, Q = 0, inf = LLONG_MAX; d y = x;
  for (;;) {
    ll lim = min(P ? (N-LP) / P : inf, Q ? (N-LQ) / Q : inf),
       a = (ll)floor(y), b = min(a, lim),
       NP = b*P + LP, NQ = b*Q + LQ;
    if (a > b) {
      // If b > a/2, we have a semi-convergent that gives us a
      // better approximation; if b = a/2, we *may* have one.
      // Return {P, Q} here for a more canonical approximation.
      return (abs(x - (d)NP / (d)NQ) < abs(x - (d)P / (d)Q)) ?
        make_pair(NP, NQ) : make_pair(P, Q);
    }
    if (abs(y = 1/(y - (d)a)) > 3 * N) {
      return {NP, NQ};
    }
    LP = P; P = NP;
    LQ = Q; Q = NQ;
  }
}
```

FracBinarySearch.h

Description: Given f and N, finds the smallest fraction $p/q \in [0, 1]$ such that $f(p/q)$ is true, and $p, q \le N$. You may want to throw an exception from f if it finds an exact solution, in which case N can be removed.
Usage: `fracBS([](Frac f) { return f.p>=3*f.q; }, 10); // {1,3}`
Time: $\mathcal{O}(\log(N))$

27ab3e, 25 lines

```
struct Frac { ll p, q; };

template<class F>
Frac fracBS(F f, ll N) {
  bool dir = 1, A = 1, B = 1;
  Frac lo{0, 1}, hi{1, 1}; // Set hi to 1/0 to search (0, N]
  if (f(lo)) return lo;
  assert(f(hi));
  while (A || B) {
    ll adv = 0, step = 1; // move hi if dir, else lo
    for (int si = 0; step; (step *= 2) >>= si) {
      adv += step;
      Frac mid{lo.p * adv + hi.p, lo.q * adv + hi.q};
      if (abs(mid.p) > N || mid.q > N || dir == !f(mid)) {
        adv -= step; si = 2;
      }
    }
    hi.p += lo.p * adv;
    hi.q += lo.q * adv;
    dir = !dir;
    swap(lo, hi);
    A = B; B = !!adv;
  }
  return dir ? hi : lo;
}
```

5.5 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with $m > n > 0$, $k > 0$, $m \perp n$, and either m or n even.

5.6 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a, except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

5.7 Estimates

$\sum_{d|n} d = O(n \log \log n)$.

The number of divisors of n is at most around 100 for $n < 5e4$, 250 for $n < 1e6$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

5.8 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \Leftrightarrow f(n) = \sum_{d|n} \mu(d) g(n/d)$$

Other useful formulas/forms:

$\sum_{d|n} \mu(d) = [n = 1]$ (very useful)

$g(n) = \sum_{n|d} f(d) \Leftrightarrow f(n) = \sum_{n|d} \mu(d/n) g(d)$

$g(n) = \sum_{1 \le m \le n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \le m \le n} \mu(m) g(\lfloor \frac{n}{m} \rfloor)$

Combinatorial (6)

6.1 Permutations

6.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800

n	11	12	13	14	15	16	17
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14

n	20	25	30	40	50	100	150	171
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX

IntPerm.h

Description: Permutation -> integer conversion. (Not order preserving.) Integer -> permutation can use a lookup table.
Time: $\mathcal{O}(n)$

044568, 6 lines

```
int permToInt(vi &v) {
  int use = 0, i = 0, r = 0;
  for (int x : v) r = r * ++i + __builtin_popcount(use & -(1 << x)),
    use |= 1 << x;          // (note: minus, not ~!)
  return r;
}
```

6.1.2 Cycles

Let $g_S(n)$ be the number of n-permutations whose cycle lengths all belong to the set S. Then

$$\sum_{n=0}^{\infty} g_S(n)\frac{x^n}{n!} = \exp\left(\sum_{n\in S}\frac{x^n}{n}\right)$$

6.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1)+D(n-2)) = nD(n-1)+(-1)^n = \left\lfloor\frac{n!}{e}\right\rceil$$

6.1.4 Burnside's lemma

Given a group G of symmetries and a set X, the number of elements of X *up to symmetry* equals

$$\frac{1}{|G|}\sum_{g\in G}|X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts "configurations" (of some sort) of length n, we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n}\sum_{k=0}^{n-1} f(\gcd(n,k)) = \frac{1}{n}\sum_{k|n} f(k)\phi(n/k).$$

6.2 Partitions and subsets

6.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1,\ p(n) = \sum_{k\in\mathbb{Z}\setminus\{0\}} (-1)^{k+1} p(n - k(3k-1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	~2e5	~2e8

6.2.2 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + ... + n_1 p + n_0$ and $m = m_k p^k + ... + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^{k}\binom{n_i}{m_i} \pmod p$.

6.2.3 Binomials

multinomial.h

Description: Computes $\binom{k_1+\cdots+k_n}{k_1,k_2,\ldots,k_n} = \frac{(\sum k_i)!}{k_1!k_2!\ldots k_n!}$.

ed4430, 5 lines

```
ll multinomial(vi& v) {
  ll c = 1, m = v.empty() ? 1 : v[0];
  FOR (i, 1, size(v)) F0R (j, v[i]) c = c * ++m / (j + 1);
  return c;
}
```

6.3 General purpose numbers

6.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t-1}$ (FFT-able). $B[0,\ldots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \ldots]$

Sums of powers:

$$\sum_{i=1}^{n} n^m = \frac{1}{m+1}\sum_{k=0}^{m}\binom{m+1}{k}B_k\cdot(n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^{\infty} f(x)dx - \sum_{k=1}^{\infty}\frac{B_k}{k!}f^{(k-1)}(m)$$

$$\approx \int_m^{\infty} f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

6.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),\ c(0,0) = 1$$
$$\textstyle\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\ldots(x+n-1)$$

$c(8,k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \ldots$

6.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j:s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j:s s.t. $\pi(j) \geq j$, k j:s s.t. $\pi(j) > j$.

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$

$$E(n,0) = E(n,n-1) = 1$$

$$E(n,k) = \sum_{j=0}^{k}(-1)^j\binom{n+1}{j}(k+1-j)^n$$

6.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$

$$S(n,1) = S(n,n) = 1$$

$$S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}j^n$$

6.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \ldots$. For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod p$$

6.3.6 Labeled unrooted trees

\# on n vertices: n^{n-2}
\# on k existing trees of size n_i: $n_1 n_2 \cdots n_k n^{k-2}$
\# with degrees d_i: $(n-2)!/((d_1-1)!\cdots(d_n-1)!)$

6.3.7 Catalan numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1,\ C_{n+1} = \frac{2(2n+1)}{n+2}C_n,\ C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \ldots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Graph (7)

7.0.1 Theorems

A matching M is maximum **iff** there is no M-augmenting path.

7.0.2 Hall's Marriage Theorem

$G = (L, R; E)$ has a matching covering all of L iff $|N(s)| \geq |S|$ for every $S \subseteq L$.
Corollary: Let $\delta = \max_{S\subseteq L}\left(|S| - |N(S)|\right)$. The size of the maximum matching is $\tau = |L| - \delta$.

7.0.3 Kőnig's Theorem

In bipartite graphs: **max matching = min vertex cover**, i.e. $\nu(G) = \tau(G)$. Hence max independent set size $\alpha(G) = |V| - \tau(G) = |V| - \nu(G)$.
To recover min vertex cover: From all unmatched vertices in L, BFS alternating paths (use non-matching edges $L \to R$; matching edges $R \to L$). Let Z be reachable vertices. Min vertex cover is $(L \setminus Z) \cup (R \cap Z)$.

7.0.4 Dilworth's Theorem (posets)

Run Floyd-Warshall for transitivity.
Width (max antichain size) = min number of chains in a partition.
Reduction: make bipartite graph with two copies of elements; edge $x_L - y_R$ if $x < y$. Then min chain decomposition size $= |P| - \nu$; width $= |P| - (|P| - \nu) = \nu$.

Mirsky's Theorem (dual): Height (max chain size) = min number of antichains in a partition.

7.0.5 Tutte's 1-Factor Theorem

$odd(G)$ is the number of components in G with an odd number of vertices. G has a perfect matching iff for all $S \subseteq V$, $\text{odd}(G - S) \leq |S|$. Tutte'Berge formula: $\nu(G) = \frac{1}{2} \min_{S \subseteq V} \left(|V| - \text{odd}(G - S) + |S|\right)$.

7.0.6 Edge cover

In any graph without isolated vertices, the size of the minimum edge cover (set of edges that touch every vertex) is given by:

$$\rho'(G) = |V| - \nu(G)$$

.

7.1 Math

7.1.1 Number of Spanning Trees

Create an $N \times N$ matrix `mat`, and for each edge $a \to b \in G$, do `mat[a][b]--, mat[b][b]++` (and `mat[b][a]--, mat[a][a]++` if G is undirected). Remove the ith row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

7.1.2 Erdős-Gallai theorem

A simple graph with node degrees $d_1 \geq \cdots \geq d_n$ exists iff $d_1 + \cdots + d_n$ is even and for every $k = 1 \ldots n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(d_i, k).$$

7.2 Fundamentals

BellmanFord.h

Description: Calculates shortest paths from s in a graph that might have negative edge weights. Unreachable nodes get dist = inf; nodes reachable through negative-weight cycles get dist = -inf. Assumes $V^2 \max |w_i| < \sim 2^{63}$.
Time: $\mathcal{O}(VE)$

f021ee, 23 lines

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; }};
struct Node { ll dist = inf; int prev = -1; };

void bellmanFord(vt<Node> &nodes, vt<Ed> &eds, int s) {
  nodes[s].dist = 0;
  sort(all(eds), [] (Ed a, Ed b) { return a.s() < b.s(); });

  int lim = size(nodes) / 2 + 2; // /3 + 100 with shuffled vertices
  FOR (i, lim) for (Ed ed : eds) {
    Node cur = nodes[ed.a], &dest = nodes[ed.b];
    if (abs(cur.dist) == inf) continue;
    ll d = cur.dist + ed.w;
    if (d < dest.dist) {
      dest.prev = ed.a;
      dest.dist = (i < lim-1 ? d : -inf);
    }
  }
  FOR (i, lim) for (Ed e : eds) {
    if (nodes[e.a].dist == -inf)
      nodes[e.b].dist = -inf;
  }
}
```

FloydWarshall.h

Description: Calculates all-pairs shortest path in a directed graph that might have negative edge weights. Input is an distance matrix m, where $m[i][j] =$ `inf` if i and j are not adjacent. As output, $m[i][j]$ is set to the shortest distance between i and j, `inf` if no path, or `-inf` if the path goes through a negative-weight cycle.
Time: $\mathcal{O}(N^3)$

35be6b, 10 lines

```
const ll inf = 1ll << 62;
void floydWarshall(vt<vt<ll>>& m) {
  int n = size(m);
  FOR (i, n) m[i][i] = min(m[i][i], 0LL);
  FOR (k, n) FOR (i, n) FOR (j, n)
    if (m[i][k] != inf && m[k][j] != inf)
      m[i][j] = min(m[i][j], max(m[i][k] + m[k][j], -inf));
  FOR (k, n) if (m[k][k] < 0) FOR (i, n) FOR (j, n)
    if (m[i][k] != inf && m[k][j] != inf) m[i][j] = -inf;
}
```

7.3 Network flow

7.3.1 Reductions

Min cut reduction

Consider min cut as an instance of this problem: Given $A[N]$ and $B[N]$ (any sign) and $C[N][N]$ (non-negative), find a binary string S of length N that minimises the following:

- Pay $A[i]$ if $S[i] = 1$
- Pay $B[i]$ if $S[i] = 0$
- Pay $C[i][j]$ if $S[i] = 1$ and $S[j] = 0$.

To calculate the answer:

- Add edges i to t of capacity $B[i] +$ `inf`.
- Add edges s to i of capacity $A[i] +$ `inf`.
- Add edges i to j of capcity $C[i][j]$.
- The answer is the maximum flow in this graph minus $n \times$ `inf`.

7.3.2 Circulation & Lower Bounds

Each edge $e = (u \to v)$ has bounds $[\ell_e, u_e]$. Find flows f_e such that:

$$\ell_e \leq f_e \leq u_e, \qquad \sum f_{\text{in}} = \sum f_{\text{out}} \text{ at every node.}$$

Let $f'_e = f_e - \ell_e$, so $0 \leq f'_e \leq u_e - \ell_e$. Define node demand:

$$d[v] = \sum_{e=(v \to w)} \ell_e - \sum_{e=(u \to v)} \ell_e$$

Then residual flow must satisfy

$$\sum f'_{\text{in}} - \sum f'_{\text{out}} = d[v].$$

Build auxiliary graph:

- For each edge $(u \to v)$, add cap $u_e - \ell_e$
- Add super source S, super sink T
- If $d[v] > 0$: add $v \to T$ cap $d[v]$
- If $d[v] < 0$: add $S \to v$ cap $-d[v]$

Let $D = \sum_{d[v]>0} d[v]$. Feasible circulation iff max $S \to T$ flow equals D.

To recover original flow, use $f_e = f'_e + \ell_e$. We can model max $s \to t$ flow with lower bounds by adding edge (t, s) and binary searching on its lower bound.

FordFulkerson.h

Description: Short algo for computing maximum flows with a bounded answer.
Time: $\mathcal{O}(FM)$

86c231, 16 lines

```
const int mx = 2000;
int seen[mx], tim;
unordered_map<int, int> adj[mx];

int flow(int s, int t) {
  auto dfs = [&] (auto &&self, int u) {
    if (u == t) return 1;
    if (exchange(seen[u], tim) == tim) return 0;
    for (auto &[v, c] : adj[u])
      if (c && self(self, v)) return --adj[u][v], ++adj[v][u];
    return 0;
  };
  int flow = 0;
  while (tim++, dfs(dfs, s)) flow++;
  return flow;
}
```

Dinic.h

Description: Flow algorithm with complexity $O(VE \log U)$ where $U = \max |\text{cap}|$. $O(\min(E^{1/2}, V^{2/3})E)$ if $U = 1$; $O(\sqrt{V}E)$ for bipartite matching.

84ac12, 48 lines

```
struct Dinic {
  struct Edge {
    int to, rev;
    ll c, oc;
    // ll flow() { return max(oc - c, 0LL); } // if you need flows
  };
  vi lvl, ptr, q;
  vt<vt<Edge>> adj;

  void init(int n) {
    lvl = ptr = q = vi(n);
    adj.resize(n);
  }

  void ae(int a, int b, ll c, ll rcap = 0) {
    adj[a].pb({b, size(adj[b]), c, c});
    adj[b].pb({a, size(adj[a]) - 1, rcap, rcap});
  }
  ll dfs(int v, int t, ll f) {
    if (v == t || !f) return f;
    for (int& i = ptr[v]; i < size(adj[v]); i++) {
      auto &[to, rev, c, _] = adj[v][i];
      if (lvl[to] == lvl[v] + 1) {
        if (ll p = dfs(to, t, min(f, c))) {
          c -= p, adj[to][rev].c += p;
          return p;
        }
      }
    }
```

```
    return 0;
  }
  ll calc(int s, int t) {
    ll flow = 0; q[0] = s;
    F0R (L, 31) do { // 'int L = 30' maybe faster for random data
      lvl = ptr = vi(size(q));
      int qi = 0, qe = lvl[s] = 1;
      while (qi < qe && !lvl[t]) {
        int v = q[qi++];
        for (Edge e : adj[v])
          if (!lvl[e.to] && e.c >> (30 - L))
            q[qe++] = e.to, lvl[e.to] = lvl[v] + 1;
      }
      while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
    } while (lvl[t]);
    return flow;
  }
  // bool left_of_min_cut(int a) { return lvl[a] != 0; }
};
```

MinCostMaxFlow.h

Description: Min-cost max-flow. If costs can be negative, call setpi before maxflow, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.
Time: $\mathcal{O}(FE\log(V))$ where F is max flow. $\mathcal{O}(VE)$ for setpi.

9414d2, 79 lines

```
#include <bits/extc++.h>

struct MCMF {
  struct edge {
    int from, to, rev;
    ll cap, cost, flow;
  };
  int n;
  vt<vt<edge>> ed;
  vt<int> seen;
  vt<ll> dist, pi;
  vt<edge*> par;

  MCMF(int n) : n(n), ed(n), seen(n), dist(n), pi(n), par(n) {}

  void ae(int from, int to, ll cap, ll cost) {
    if (from == to) return;
    ed[from].push_back({from, to, size(ed[to]),
      cap, cost, 0});
    ed[to].push_back({ to,  from, size(ed[from]) - 1,
      0, -cost, 0});
  }

  void path(int s) {
    fill(all(seen), 0);
    fill(all(dist), INF);
    dist[s] = 0; ll di;

    __gnu_pbds::priority_queue<pair<ll, int>> q;
    vt<decltype(q)::point_iterator> its(n);
    q.push({ 0, s });

    while (!q.empty()) {
      s = q.top().second; q.pop();
      seen[s] = 1; di = dist[s] + pi[s];
      for (edge& e : ed[s]) if (!seen[e.to]) {
        ll val = di - pi[e.to] + e.cost;
        if (e.cap - e.flow > 0 && val < dist[e.to]) {
          dist[e.to] = val;
          par[e.to] = &e;
          if (its[e.to] == q.end())
            its[e.to] = q.push({ -dist[e.to], e.to });
          else
            q.modify(its[e.to], { -dist[e.to], e.to });
        }
      }
    }
    F0R (i, n) pi[i] = min(pi[i] + dist[i], INF);
  }

  pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s), seen[t]) {
      ll fl = INF;
      for (edge* x = par[t]; x; x = par[x->from])
        fl = min(fl, x->cap - x->flow);

      totflow += fl;
      for (edge* x = par[t]; x; x = par[x->from]) {
        x->flow += fl;
        ed[x->to][x->rev].flow -= fl;
      }
    }
    FOR (i, n) for(edge& e : ed[i]) totcost += e.cost * e.flow;
    return {totflow, totcost/2};
  }

  // If some costs can be negative, call this before maxflow:
  // void setpi(int s) { // (otherwise, leave this out)
  //   fill(all(pi), INF); pi[s] = 0;
  //   int it = n, ch = 1; ll v;
  //   while (ch-- && it--)
  //     FOR (i, n) if (pi[i] != INF)
  //       for (edge& e : ed[i]) if (e.cap)
  //         if ((v = pi[i] + e.cost) < pi[e.to])
  //           pi[e.to] = v, ch = 1;
  //   assert(it >= 0); // negative cost cycle
  // }
};
```

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.
Time: $\mathcal{O}(V^3)$

d4d36f, 21 lines

```
pair<int, vt<int>> globalMinCut(vt<vt<int>> mat) {
  pair<int, vt<int>> best = {INT_MAX, {}};
  int n = size(mat);
  vt<vt<int>> co(n);
  F0R (i, n) co[i] = {i};
  FOR(ph, 1, n) {
    vt<int> w = mat[0];
    size_t s = 0, t = 0;
    F0R (it, n - ph) { // O(V^2) -> O(E log V) with prio. queue
      w[t] = INT_MIN;
      s = t, t = max_element(all(w)) - w.begin();
      F0R (i, n) w[i] += mat[t][i];
    }
    best = min(best, {w[t] - mat[t][t], co[t]});
    co[s].insert(co[s].end(), all(co[t]));
    F0R (i, n) mat[s][i] += mat[t][i];
    F0R (i, n) mat[i][s] = mat[s][i];
    mat[0][t] = INT_MIN;
  }
  return best;
}
```

NetworkSimplex.h

Description: Solves minimum cost circulation problem. To convert to a "normal" mcmf, add an edge from t -> s of big capacity and negative inf cost (remember to add this back on to answer!). If you don't necessarily need to maximise flow, add free edge from s -> t. Edge i (one indexed) is ns.edges[2 * i]]. Works with negative cost cycles.
Time: $\mathcal{O}(VE)$ on average maybe

c6f865, 93 lines

```
struct NetworkSimplex {
  using Flow = int;
  using Cost = ll;
  struct Edge {
    int nxt, to;
    Flow cap;
    Cost cost;
  };
  vector<Edge> edges;
  vector<int> head, fa, fe, mark, cyc;
  vector<Cost> dual;
  int ti;

  NetworkSimplex(int n)
      : head(n, 0), fa(n), fe(n), mark(n), cyc(n + 1), dual(n), ti(0)
          {
    edges.push_back({0, 0, 0, 0});
    edges.push_back({0, 0, 0, 0});
  }

  int ae(int u, int v, Flow cap, Cost cost) {
    assert(size(edges) % 2 == 0);
    int e = size(edges);
    edges.push_back({head[u], v, cap, cost});
    head[u] = e;
    edges.push_back({head[v], u, 0, -cost});
    head[v] = e + 1;
    return e;
  }

  void init_tree(int x) {
    mark[x] = 1;
    for (int i = head[x]; i; i = edges[i].nxt) {
      int v = edges[i].to;
      if (!mark[v] and edges[i].cap) {
        fa[v] = x, fe[v] = i;
        init_tree(v);
      }
    }
  }

  Cost phi(int x) {
    if (mark[x] == ti) return dual[x];
    return mark[x] = ti, dual[x] = phi(fa[x]) - edges[fe[x]].cost;
  }

  void push_flow(int e, Cost &cost) {
    int pen = edges[e ^ 1].to, lca = edges[e].to;
    ti++;
    while (pen) mark[pen] = ti, pen = fa[pen];
    while (mark[lca] != ti) mark[lca] = ti, lca = fa[lca];

    int e2 = 0, f = edges[e].cap, path = 2, clen = 0;
    for (int i = edges[e ^ 1].to; i != lca; i = fa[i]) {
      cyc[++clen] = fe[i];
      if (edges[fe[i]].cap < f)
        f = edges[fe[e2 = i] ^ (path = 0)].cap;
    }
    for (int i = edges[e].to; i != lca; i = fa[i]) {
      cyc[++clen] = fe[i] ^ 1;
      if (edges[fe[i] ^ 1].cap <= f)
        f = edges[fe[e2 = i] ^ (path = 1)].cap;
```

```
    }
    cyc[++clen] = e;

    for (int i = 1; i <= clen; ++i) {
      edges[cyc[i]].cap -= f, edges[cyc[i] ^ 1].cap += f;
      cost += edges[cyc[i]].cost * f;
    }
    if (path == 2)  return;

    int laste = e ^ path, last = edges[laste].to, cur = edges[laste ^
         1].to;
    while (last != e2) {
      mark[cur]--;
      laste ^= 1;
      swap(laste, fe[cur]);
      swap(last, fa[cur]); swap(last, cur);
    }
  }

  Cost compute() {
    Cost cost = 0;
    init_tree(0);
    mark[0] = ti = 2, fa[0] = cost = 0;
    int ncnt = size(edges) - 1;
    for (int i = 2, pre = ncnt; i != pre; i = i == ncnt ? 2 : i + 1)
      if (edges[i].cap and edges[i].cost < phi(edges[i ^ 1].to) - phi(
           edges[i].to))
        push_flow(pre = i, cost);
    ti++;
    for (int u = 0; u < size(dual); ++u)
      phi(u);
    return cost;
  }
};
```

7.4 Matching

DFSMatching.h

Description: Simple bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.
Usage: `vi btoa(m, -1); dfsMatching(g, btoa);`
Time: $\mathcal{O}(VE)$

c2ed08, 22 lines

```
bool find(int j, vt<vi> &g, vi &btoa, vi &vis) {
  if (btoa[j] == -1) return 1;
  vis[j] = 1; int di = btoa[j];
  for (int e : g[di])
    if (!vis[e] && find(e, g, btoa, vis)) {
      btoa[e] = di;
      return 1;
    }
  return 0;
}
int dfsMatching(vt<vi> &g, vi &btoa) {
  vi vis;
  FOR (i, size(g)) {
    vis.assign(size(btoa), 0);
    for (int j : g[i])
      if (find(j, g, btoa, vis)) {
        btoa[j] = i;
        break;
      }
  }
  return size(btoa) - (int) count(all(btoa), -1);
}
```

MinimumVertexCover.h

Description: Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

"DFSMatching.h" 6cc1d8, 20 lines

```
vi cover(vector<vi>& g, int n, int m) {
  vi match(m, -1);
  int res = dfsMatching(g, match);
  vt<bool> lfound(n, true), seen(m);
  for (int it : match) if (it != -1) lfound[it] = false;
  vi q, cover;
  FOR (i, n) if (lfound[i]) q.push_back(i);
  while (!q.empty()) {
    int i = q.back(); q.pop_back();
    lfound[i] = 1;
    for (int e : g[i]) if (!seen[e] && match[e] != -1) {
      seen[e] = true;
      q.pb(match[e]);
    }
  }
  FOR (i, n) if (!lfound[i]) cover.push_back(i);
  FOR (i, m) if (seen[i]) cover.push_back(n+i);
  assert(size(cover) == res);
  return cover;
}
```

hopcroftKarp.h

Description: Fast incremental bipartite matching. Zero-indexed.
Usage: `{operator[]}` for the pair of right node i, `{n}` is the size of the rhs, `{add(v)}` to add adjacency list of node on lhs
Time: $\mathcal{O}\left(\sqrt{V}E\right)$

fc3e44, 40 lines

```
struct Matching : vi {
  vt<vi> adj;
  vi rank, low, pos, vis, seen;
  int k{0};
  // n = size of rhs
  Matching(int n) : vi(n, -1), rank(n) {}
  bool add(vi vec) {
    adj.pb(std::move(vec));
    low.pb(0), pos.pb(0), vis.pb(0);
    if (size(adj.back())) {
      int i = k;
    nxt:
      seen.clear();
      if (dfs(size(adj)-1, ++k-i)) return 1;
      for (auto &v : seen) for (auto &e : adj[v]) {
        if (rank[e] < inf && vis[at(e)] < k) goto nxt;
      }
      for (auto &v : seen) {
        low[v] = inf;
        for (auto &w : adj[v]) rank[w] = inf;
      }
    }
    return 0;
  }
  bool dfs(int v, int g) {
    if (vis[v] < k) vis[v] = k, seen.pb(v);
    while (low[v] < g) {
      int e = adj[v][pos[v]];
      if (at(e) != v && low[v] == rank[e]) {
        rank[e]++;
        if (at(e) == -1 || dfs(at(e), rank[e])) {
          return at(e) = v, 1;
        }
      } else if (++pos[v] == size(adj[v])) {
        pos[v] = 0; low[v]++;
      }
    }
    return 0;
  }
};
```

WeightedMatching.h

Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires $N \leq M$.
Time: $\mathcal{O}\left(N^2M\right)$

2e27e7, 31 lines

```
pair<int, vi> hungarian(const vt<vi> &a) {
  if (a.empty()) return {0, {}};
  int n = size(a) + 1, m = size(a[0]) + 1;
  vi u(n), v(m), p(m), ans(n - 1);
  FOR (i, 1, n) {
    p[0] = i;
    int j0 = 0;
    vi dist(m, INT_MAX), pre(m, -1);
    vector<bool> done(m + 1);
    do {
      done[j0] = true;
      int i0 = p[j0], j1, delta = INT_MAX;
      FOR (j, 1, m) if (!done[j]) {
        auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
        if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
        if (dist[j] < delta) delta = dist[j], j1 = j;
      }
      FOR (j, m) {
        if (done[j]) u[p[j]] += delta, v[j] -= delta;
        else dist[j] -= delta;
      }
      j0 = j1;
    } while (p[j0]);
    while (j0) { // update alternating path
      int j1 = pre[j0];
      p[j0] = p[j1], j0 = j1;
    }
  }
  FOR (j, 1, m) if (p[j]) ans[p[j] - 1] = j - 1;
  return {-v[0], ans}; // min cost
}
```

Blossom.h

Description: Matching for general graphs. 1-indexed!
Time: $\mathcal{O}(NM)$

0b8afb, 59 lines

```
// 1-indexed!
struct Blossom {
  int n, h, t, cnt;
  vpi edges;
  vi vis, q, mate, col, fa, pre, he;
  void ae(int u, int v) {
    assert(u && v);
    edges.pb({he[u], v}); he[u] = size(edges) - 1;
    edges.pb({he[v], u}); he[v] = size(edges) - 1;
  }
  inline int get(int u) { return fa[u] == u ? u : fa[u] = get(fa[u]);
       }
  void aug(int u, int v) {
    for (int p; u; u = p, v = pre[p])
      p = mate[v], mate[mate[u] = v] = u;
  }
  void init(int _n) {
    n = _n;
    vis = q = mate = col = fa = pre = he = vi(n + 1);
  }
  int lca(int u, int v) {
```

```
    for (cnt++;; u = pre[mate[u]]) {
      if (v) swap(u, v);
      if (vis[u = get(u)] == cnt) return u;
      vis[u] = cnt;
    }
  }
  void blo(int u, int v, int f) {
    for (int p; get(u) != f; v = p, u = pre[p]) {
      p = mate[u]; pre[u] = v; fa[u] = fa[p] = f;
      if (col[p] != 1) col[q[++t] = p] = 1;
    }
  }
  bool bfs(int u) {
    FOR (i, 1, n + 1) col[i] = 0, fa[i] = i;
    h = 0; q[t = 1] = u; col[u] = 1;
    while (h != t) {
      int x = q[++h];
      for (int i = he[x]; i; i = edges[i].f) {
        int y = edges[i].s;
        if (!col[y]) {
          if (!mate[y]) { aug(y, x); return 1; }
          pre[y] = x;
          col[y] = 2;
          col[q[++t] = mate[y]] = 1;
        } else if (col[y] == 1 && get(x) != get(y)) {
          int p = lca(x, y);
          blo(x, y, p);
          blo(y, x, p);
        }
      }
    }
    return 0;
  }
  int solve() {
    int ans = 0;
    FOR (i, 1, n + 1) if (!mate[i]) ans += bfs(i);
    return ans;
  }
};
```

7.5 DFS algorithms

SCC2.h

Description: Finds strongly connected components in a directed graph. comps has edges right to left.
Time: $\mathcal{O}(E+V)$

94cd75, 21 lines

```
using G = vt<vi>; // vt<basic_string<int>> faster
struct SCC {
  int ti = 0; G &adj;
  vi disc, comp, stk;
  SCC(int n, G &adj) : adj(adj), disc(n), comp(n, -1) {
    FOR (i, n) if (!disc[i]) dfs(i);
    // reverse(all(comps));
  }
  int dfs(int u) {
    int low = disc[u] = ++ti;
    stk.pb(u);
    for (int v : adj[u]) if (comp[v] == -1)
      low = min(low, disc[v] ?: dfs(v));
    if (low == disc[u]) {
      // comps.pb(u);
      for (int y = -1; y != u;)
        comp[y = stk.back()] = u, stk.pop_back();
    }
    return low;
  }
};
```

ArticulationPoints.h

Description: Finds all articulation points in a graph.
Time: $\mathcal{O}(E+V)$

707c6d, 21 lines

```
struct Arts : vi {
  int t;
  vi tin;
  Arts(int n, vt<vi> &adj) : vi(n), t(0), tin(n) {
    FOR (u, n) if (!tin[u]) dfs(u, adj, 0);
  }

  int dfs(int u, vt<vi> &adj, int cnt = 1) {
    int dp = tin[u] = ++t;
    for (int v : adj[u]) {
      if (tin[v]) dp = min(dp, tin[v]);
      else {
        int up = dfs(v, adj);
        dp = min(dp, up);
        cnt += up >= tin[u];
      }
    }
    at(u) = cnt >= 2;
    return dp;
  }
};
```

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph. In a biconnected component there are at least two distinct paths between any two nodes (a cycle exists through them). Note that a node can be in several components. An edge which is not in a component is a bridge, i.e., not part of any cycle. Note that degree 0 nodes are not considered components.
Time: $\mathcal{O}(E+V)$

061a2d, 47 lines

```
struct BCC {
  int t;
  vt<vt<pi>> adj;
  vt<vi> comps; // lists of edges of bcc
  vi tin, stk, is_bridge; // stk for bcc only
  // vi is_art;

  void init(int n, vt<pi> &edges) {
    int m = size(edges);
    adj.resize(n);
    FOR (i, m) {
      auto [u, v] = edges[i];
      adj[u].pb({v, i});
      adj[v].pb({u, i});
    }
    t = 0;
    tin.resize(n);
    // is_art.resize(n);
    is_bridge.resize(m);
    FOR (u, n) if (!tin[u]) dfs(u, -1);
    // if we include bridges as 2-node bcc
    // FOR (i, m) if (is_bridge[i]) comps.pb({i});
  }

  int dfs(int u, int par) {
    int me = tin[u] = ++t, dp = me;
    // int cnt = (par != -1); // art
    for (auto [v, ei] : adj[u]) if (ei != par) {
      if (tin[v]) {
        dp = min(dp, tin[v]);
        if (tin[v] < me) stk.pb(ei); // bcc
      } else {
        int si = size(stk), up = dfs(v, ei);
        dp = min(dp, up);
        // cnt += up >= me; // art
        if (up == me) { // bcc
          stk.pb(ei);
          comps.pb({si + all(stk)});
          stk.resize(si);
        } else if (up < me) stk.push_back(ei); // bcc
        if (up > me) is_bridge[ei] = 1; // bridges
      }
    }
    // is_art[u] = cnt >= 2; // art
    return dp;
  }
};
```

BlockCutTree.h

Description: First, use BiconnectedComponents to locate VERTEX components (including degree 0 nodes). To build a block cut tree, make a bipartite graph: Put all the normal nodes on the left, and make a new node for each bcc on the right. Draw edges from normal nodes to BCC that contain them. Note that the graph may be disconnected.

2sat.h

Description: Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type $(a||b)\&\&(!a||c)\&\&(d||!b)\&\&\ldots$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).
Usage: TwoSat ts(number of boolean variables);
ts.either(0, ~3); // var 0 is true or var 3 is false
ts.force(2); // var 2 is true
ts.at_most_one({0,~1,2}); // <= 1 of vars 0, ~1 and 2 are true
ts.solve(); // returns true iff it is solvable
ts.values[0..N-1] holds the assigned values to the vars
Time: $\mathcal{O}(N+E)$, where N is the number of boolean variables, and E is the number of clauses.

903fd1, 52 lines

```
// using G = vt<basic_string<int>>;
using pi = pair<int, int>;
struct TwoSAT {
  int n;
  vt<pi> edges;
  // TwoSat sat{n};
  int add() { return n++; }
  void either(int x, int y) { // x | y
    x = max(2 * x, -1 - 2 * x); // ~(2 * x)
    y = max(2 * y, -1 - 2 * y); // ~(2 * y)
    edges.pb({x, y});
  }
  void implies(int x, int y) { either(~x, y); }
  void force(int x) { either(x, x); }
  void exactly_one(int x, int y) {
    either(x, y), either(~x, ~y);
  }
  void tie(int x, int y) {
    implies(x, y), implies(~x, ~y);
  }
  void nand(int x, int y ) { either(~x, ~y); }
  void at_most_one(const vi &li) {
    if (size(li) <= 1) return;
    int cur = ~li[0];
    FOR (i, 2, size(li)) {
      int next = add();
      either(cur, ~li[i]);
      either(cur,next);
      either(~li[i], next);
      cur = ~next;
    }
    either(cur, ~li[1]);
  }
  vt<bool> solve() {
    G adj(2 * n);
```

```
    for(auto& e : edges) {
      adj[e.f ^ 1].pb(e.s);
      adj[e.s ^ 1].pb(e.f);
    }
    SCC scc(2 * n, adj);
    reverse(all(scc.comps)); // reverse topo order
    for (int i = 0; i < 2 * n; i += 2)
      if (scc.comp[i] == scc.comp[i ^ 1]) return {};
    vi tmp(2 * n);
    for (auto i : scc.comps) {
      if (!tmp[i]) tmp[i] = 1, tmp[scc.comp[i ^ 1]] = -1;
    }
    vt<bool> ans(n);
    F0R (i, n) ans[i] = tmp[scc.comp[2 * i]] == 1;
    return ans;
  }
};
```

EulerWalk.h

Description: Eulerian undirected/directed path/cycle algorithm. For edges, push the edge u came from instead of current node. First, the graph (after removing directivity) must be connected. For undirected graphs, a tour exists when all nodes have even degree. For directed graphs, a tour exists when all nodes have equal in and out degree. For trails, the condition is the same as if you added an edge from t -> s.
Time: $\mathcal{O}(V+E)$

378922, 14 lines

```
int n, m;
vt<vt<pair<int, int>>> adj;
vt<int> ret, used;

//
void dfs(int u) {
  while (adj[u].size()) {
    auto [v, ei] = adj[u].back();
    adj[u].pop_back();
    if (used[ei]++) continue;
    dfs(v);
  }
  ret.push_back(u);
}
```

7.6 Heuristics

MaximumClique.h

Description: Quickly finds a maximum clique of a graph (given as symmetric bitset matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.
Time: Runs in about 1s for n=155 and worst case random graphs (p=.90). Runs faster for sparse graphs.

f7c0bc, 49 lines

```
typedef vector<bitset<200>> vb;
struct Maxclique {
  double limit=0.025, pk=0;
  struct Vertex { int i, d=0; };
  typedef vector<Vertex> vv;
  vb e;
  vv V;
  vector<vi> C;
  vi qmax, q, S, old;
  void init(vv& r) {
    for (auto& v : r) v.d = 0;
    for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
    sort(all(r), [](auto a, auto b) { return a.d > b.d; });
    int mxD = r[0].d;
    rep(i,0,sz(r)) r[i].d = min(i, mxD) + 1;
  }
  void expand(vv& R, int lev = 1) {
    S[lev] += S[lev - 1] - old[lev];
    old[lev] = S[lev - 1];
    while (sz(R)) {
      if (sz(q) + R.back().d <= sz(qmax)) return;
      q.push_back(R.back().i);
      vv T;
      for(auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
      if (sz(T)) {
        if (S[lev]++ / ++pk < limit) init(T);
        int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
        C[1].clear(), C[2].clear();
        for (auto v : T) {
          int k = 1;
          auto f = [&](int i) { return e[v.i][i]; };
          while (any_of(all(C[k]), f)) k++;
          if (k > mxk) mxk = k, C[mxk + 1].clear();
          if (k < mnk) T[j++].i = v.i;
          C[k].push_back(v.i);
        }
        if (j > 0) T[j - 1].d = 0;
        rep(k,mnk,mxk + 1) for (int i : C[k])
          T[j].i = i, T[j++].d = k;
        expand(T, lev + 1);
      } else if (sz(q) > sz(qmax)) qmax = q;
      q.pop_back(), R.pop_back();
    }
  }
  vi maxClique() { init(V), expand(V); return qmax; }
  Maxclique(vb conn) : e(conn), C(sz(e)+1), S(sz(C)), old(S) {
    rep(i,0,sz(e)) V.push_back({i});
  }
};
```

7.7 Trees

BinaryLifting.h

Description: Calculate power of two jumps in a tree, to support fast upward jumps and LCAs. Assumes the root node points to itself.
Time: construction $\mathcal{O}(N \log N)$, queries $\mathcal{O}(\log N)$

9fc465, 25 lines

```
vt<vi> build_table(vi &par){
  int d = 1;
  while ((1 << d) < size(par)) d++;
  vt<vt<int>> jmp(d, par);
  FOR (i, 1, d) FOR (j, 0, size(par))
    jmp[i][j] = jmp[i - 1][jmp[i - 1][j]];
  return jmp;
}

int jump(vt<vi>& jmp, int u, int d){
  F0R (i, size(jmp))
    if (1 & d >> i) u = jmp[i][u];
  return u;
}

int lca(vt<vi> &jmp, vi &depth, int a, int b) {
  if (depth[a] < depth[b]) swap(a, b);
  a = jump(jmp, a, depth[a] - depth[b]);
  if (a == b) return a;
  for (int i = size(jmp); i--;) {
    int c = jmp[i][a], d = jmp[i][b];
    if (c != d) a = c, b = d;
  }
  return jmp[0][a];
}
```

LCA.h

Description: Data structure for computing lowest common ancestors in a tree (with 0 as root). C should be an adjacency list of the tree, either directed or undirected.
Time: $\mathcal{O}(N \log N + Q)$

"../data-structures/RMQ.h" bf464a, 20 lines

```
struct LCA {
  int t = 0;
  vi time, path, ret;
  RMQ<int> rmq;

  LCA(vt<vi>& adj) : time(size(adj)), rmq((dfs(0, -1, adj), ret)) {}
  void dfs(int u, int p, vt<vi> &adj) {
    time[u] = t++;
    for (int v : adj[u]) if (v != p) {
      path.push_back(u), ret.push_back(time[u]);
      dfs(v, u, adj);
    }
  }

  int operator()(int u, int v) {
    if (u == v) return u;
    tie(u, v) = minmax(time[u], time[v]);
    return path[rmq.query(u, v)];
  }
};
```

VirtualTree.h

Description: Computes virtual tree. pos is inorder dfs time, and returns pairs of (par, child).
Time: $\mathcal{O}(|S| \log |S|)$

"LCA.h" 4ff13b, 14 lines

```
// pos is dfs time
// pairs of {ancestor, child}
vt<pl> virtual_tree(vt<ll>& nodes) {
  auto cmp = [&] (ll u, ll v) { return pos[u] < pos[v]; };
  sort(all(nodes), cmp);
  int sz = size(nodes);
  FOR (i, sz - 1) nodes.pb(lca(nodes[i], nodes[i + 1]));
  sort(all(nodes), cmp);
  nodes.erase(unique(all(nodes)), nodes.end());
  vt<pl> res;
  FOR (i, (int) size(nodes) - 1)
  res.pb({lca(nodes[i], nodes[i + 1]), nodes[i + 1]});
  return res;
}
```

HLD.h

Description: Decomposes a tree into vertex disjoint heavy paths and light edges such that the path from any leaf to the root contains at most log(n) light edges. Code does additive modifications and max queries, but can support commutative segtree modifications/queries on paths and subtrees. Takes as input the full adjacency list. in_edges being true means that values are stored in the edges, as opposed to the nodes. All values initialized to the segtree default. Root must be 0.
Time: $\mathcal{O}\left((\log N)^2\right)$

7b3675, 40 lines

```
template<bool in_edges> struct HLD {
  int n;
  vt<vi> adj;
  vi par, root, sz, pos;
  HLD(vt<vi> &adj) : n(size(adj)), adj(adj), par(n), root(n), sz(n),
      pos(n) {
    dfs_sz(0);
    dfs_hld(0);
  }
  void dfs_sz(int u) {
    sz[u] = 1;
    for (int& v : adj[u]) {
      par[v] = u;
      adj[v].erase(find(all(adj[v]), u));
      dfs_sz(v);
```

```
      sz[u] += sz[v];
      if (sz[v] > sz[adj[u][0]]) swap(v, adj[u][0]);
    }
  }
  void dfs_hld(int u) {
    pos[u] = time++;
    for (int& v : adj[u]) {
      root[v] = (v == adj[u][0] ? root[u] : v);
      dfs_hld(v);
    }
  }
  void init(int _n) {
    n = _n;
    adj.resize(n);
    par = root = sz = pos = vi(n);
  }
  template <class Op>
  void process(int u, int v, Op op) {
    for (;; v = par[root[v]]) {
      if (pos[u] > pos[v]) swap(u, v);
      if (root[u] == root[v]) break;
      op(pos[root[v]], pos[v] + 1);
    }
    op(pos[u] + in_edges, pos[v] + 1); // u is lca
  }
};
```

CentroidDecomp.h

Description: Give order to call dfs on.

Time: $\mathcal{O}(N)$

2223b0, 14 lines

```
vector<pair<int, int>> ord;
auto dfs1 = [&] (auto &&self, int u, int p) -> int {
  int msk1 = 0, msk2 = 0;
  for (int v : adj[u]) if (v - p) {
    int res = self(self, v, u);
    msk2 |= msk1 & res;
    msk1 |= res;
  }
  int res = (msk1 | ((1 << __lg(2 * msk2 + 1)) - 1)) + 1;
  ord.push_back({-__builtin_ctz(res), u});
  return res;
};
dfs1(dfs1, 0, -1);
sort(all(ord));
```

7.8 Other

MatroidIntersection.h

Description: Given two matroids, finds the largest common independent set. For the color and graph matroids, this would be the largest forest where no two edges are the same color. A matroid has 3 functions - check(int x): returns if current matroid can add x without becoming dependent - add(int x): adds an element to the matroid (guaranteed to never make it dependent) - clear(): sets the matroid to the empty matroid The matroid is given an int representing the element, and is expected to convert it (e.g: the color or the endpoints) Pass the matroid with more expensive add/clear operations to M1.

Time: $R^2N(M2.add + M1.check + M2.check) + R^3M1.add + R^2M1.clear + RNM2.clear$

"../data-structures/UnionFind.h" 9812a7, 60 lines

```
struct ColorMat {
  vi cnt, clr;
  ColorMat(int n, vector<int> clr) : cnt(n), clr(clr) {}
  bool check(int x) { return !cnt[clr[x]]; }
  void add(int x) { cnt[clr[x]]++; }
  void clear() { fill(all(cnt), 0); }
};
struct GraphMat {
  UF uf;
  vector<array<int, 2>> e;
  GraphMat(int n, vector<array<int, 2>> e) : uf(n), e(e) {}
  bool check(int x) { return !uf.sameSet(e[x][0], e[x][1]); }
  void add(int x) { uf.join(e[x][0], e[x][1]); }
  void clear() { uf = UF(sz(uf.e)); }
};
template <class M1, class M2> struct MatroidIsect {
  int n;
  vector<char> iset;
  M1 m1; M2 m2;
  MatroidIsect(M1 m1, M2 m2, int n) : n(n), iset(n + 1), m1(m1), m2(m2
      ) {}
  vi solve() {
    rep(i,0,n) if (m1.check(i) && m2.check(i))
      iset[i] = true, m1.add(i), m2.add(i);
    while (augment());
    vi ans;
    rep(i,0,n) if (iset[i]) ans.push_back(i);
    return ans;
  }
  bool augment() {
    vector<int> frm(n, -1);
    queue<int> q({n}); // starts at dummy node
    auto fwdE = [&](int a) {
      vi ans;
      m1.clear();
      rep(v, 0, n) if (iset[v] && v != a) m1.add(v);
      rep(b, 0, n) if (!iset[b] && frm[b] == -1 && m1.check(b))
        ans.push_back(b), frm[b] = a;
      return ans;
    };
    auto backE = [&](int b) {
      m2.clear();
      rep(cas, 0, 2) rep(v, 0, n)
        if ((v == b || iset[v]) && (frm[v] == -1) == cas) {
          if (!m2.check(v))
            return cas ? q.push(v), frm[v] = b, v : -1;
          m2.add(v);
        }
      return n;
    };
    while (!q.empty()) {
      int a = q.front(), c; q.pop();
      for (int b : fwdE(a))
        while((c = backE(b)) >= 0) if (c == n) {
          while (b != n) iset[b] ^= 1, b = frm[b];
          return true;
        }
    }
    return false;
  }
};
```

Geometry (8)

8.1 Notes

For checking imprecise equalities use `abs(x) < eps`.
Clamp values to $[-1, 1]$ for `acos`, etc.

8.2 Precision

double can precisely store integers up to 9×10^{15}, and **long double** can store up to 1.8×10^{19}.

Operations $(+, \times, -, /, \sqrt{x})$ will be rounded to the nearest representable value. Hence, the **relative** error on the result of one of these operations is bounded by $\epsilon = 1.2 \times 10^{-16}$ for **double** and $\epsilon = 5.5 \times 10^{-20}$ for **long double**.
If computations are precise up to a relative error of ϵ, and the magnitude of our values never exceeds M, then the absolute error of an operation is at most $M\epsilon$.
It is typically **more useful to consider absolute error** instead of relative error (1).
A series of n $+$ and $-$ operations result in an absolute error of at most $nM\epsilon$.
With $\times$, the absolute error of a d-dimensional value computed in n operations is $M^d((1+\epsilon)^n - 1) \approx nM^d\epsilon$ (excluding multiplication by adimensional values).
Imprecisions on x in $\frac{1}{x}$ can cause an arbitrary error (2).
Imprecisions on x in $\sqrt{x}$ are typically bad, particularly near 0. (3)
Note that $\frac{1}{x}$ and $\sqrt{x}$ perform poorly on imprecise inputs: when working with exact inputs, the `IEEE 754` standard guarantees a relative error of ϵ/absolute error of $M\epsilon$.
In particular, circle/line/circle-line intersections on exact coordinates have relative error proportional to ϵ/absolute error proportional to $M\epsilon$.

(1) subtracting two large imprecise values can blow up relative errors
(2) division by a value near 0 can result in an arbitrarily large error in both positive and negative
(3) assuming arguments to $\sqrt{x}$ are non-negative, imprecisions on x will have the most impact when x is near 0: for a given imprecision δ, the biggest imprecision on $\sqrt{x}$ it might cause is $\sqrt{\delta}$. This is quite bad: an argument with imprecision $nM^2\epsilon$ will become an imprecision of $\sqrt{n}M\sqrt{\epsilon}$. This can appear with circle intersections and computing roots to quadratics.

8.3 Common

8.3.1 Equation of line through two points

Given points p and q, equation of the line $ax + by + c = 0$ is given by:

$$a = p_y - q_y$$
$$b = q_x - p_x$$
$$c = p \times q$$

8.4 Geometric primitives

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

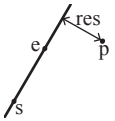
23b2a4, 34 lines

```
template <class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T>
struct Point {
  using P = Point<T>;
  T x, y;
  #define op1(o) P operator o (P p) const { return {x o p.x, y o p.y};
      }
```

```
  op1(+) op1(-)
  #define op2(o) P operator o (T z) const { return {x o z, y o z}; }
  op2(*) op2(/)
  T dot(P p) const { return x * p.x + y * p.y; }
  T cross(P p) const { return x * p.y - y * p.x; }
  #define op3(o) T o (P a, P b) const { return (a - *this). o (b - *
       this); }
  op3(dot) op3(cross)
  #define op4(o) bool operator o (P p) const { return tie(x, y) o tie(
       p.x, p.y); }
  op4(<) op4(==)
  int half() const { return y < 0 || (y == 0 && x < 0); }
  // radial
  // bool operator<(P p) const {
  //   return make_pair(half(), 0ll) < make_pair(p.half(), cross(p));
  // }
  T dist2() const { return x * x + y * y; }
  db dist() const { return sqrtl((db) dist2()); }
  // angle to x-axis in interval [-pi, pi]
  db angle() const { return atan2l(y, x); }
  P unit() const { return *this / dist(); }
  P perp() const { return {-y, x}; } // rotate 90 degrees left
  P normal() const { return perp().unit(); }
  P rotate(db a) const {
    return P(x * cos(a) - y * sin(a), x * sin(a) + y * cos(a));
  }
  friend ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.x << ", " << p.y << ") ";
  }
};
```

lineDistance.h

Description:
Returns the signed distance between point p and the line containing points a and b. Positive value on left side and negative on right as seen from a towards b. a==b gives nan. P is supposed to be Point<T> or Point3D<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using Point3D will always give a non-negative distance. For Point3D, call .dist on the result of the cross product.

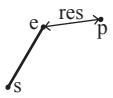


"Point.h" 3b34c4, 4 lines

```
template<class P>
double line_dist(const P& a, const P& b, const P& p) {
  return (double) (b - a).cross(p - a) / (b - a).dist();
}
```

SegmentDistance.h

Description:
Returns the shortest distance between point p and the line segment from point s to e.



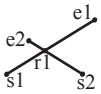
Usage: Point<double> a, b(2,2), p(1,1);
bool on_segment = seg_dist(a,b,p) < 1e-10;

"Point.h" 579797, 5 lines

```
double seg_dist(P& s, P& e, P& p) {
  if (s == e) return (p - s).dist();
  auto d = (e - s).dist2(), t = min(d, max(.0, (p - s).dot(e - s)));
  return ((p - s) * d - (e - s) * t).dist() / d;
}
```

SegmentIntersection.h

Description:
If a unique intersection point between the line segments going from s1 to e1 and from s2 to e2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.



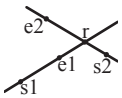
Usage: vector<P> inter = seg_inter(s1,e1,s2,e2);
if (size(inter) == 1)
cout << "segments intersect at " << inter[0] << endl;

"Point.h", "OnSegment.h" bd6e14, 13 lines

```
template<class P> vt<P> seg_inter(P a, P b, P c, P d) {
  auto oa = c.cross(d, a), ob = c.cross(d, b),
     oc = a.cross(b, c), od = a.cross(b, d);
  // Checks if intersection is single non-endpoint point.
  if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
    return {(a * ob - b * oa) / (ob - oa)};
  set<P> s;
  if (on_segment(c, d, a)) s.insert(a);
  if (on_segment(c, d, b)) s.insert(b);
  if (on_segment(a, b, c)) s.insert(c);
  if (on_segment(a, b, d)) s.insert(d);
  return {all(s)};
}
```

lineIntersection.h

Description:
If a unique intersection point of the lines going through s1,e1 and s2,e2 exists {1, point} is returned. If no intersection point exists {0, (0,0)} is returned and if infinitely many exists {-1, (0,0)} is returned. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.



Usage: auto res = line_inter(s1,e1,s2,e2);
if (res.first == 1)
cout << "intersection point at " << res.second << endl;

"Point.h" 5d933a, 17 lines

```
template<class P>
pair<int, P> line_inter(P s1, P e1, P s2, P e2) {
  auto d = (e1 - s1).cross(e2 - s2);
  if (d == 0) // if parallel
    return {-(s1.cross(e1, s2) == 0), P(0, 0)};
  auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
  return {1, (s1 * p + e1 * q) / d};
}

template<class P>
pair<int, P> line_inter(P s0, P d0, P s1, P d1) {
  auto d = (d0).cross(d1);
  if (d == 0) // if parallel
    return {0, {}};
  auto p = (s0 + d0 - s1).cross(d1), q = (d1).cross(s0 - s1);
  return {1, (s0 * p + (s0 + d0) * q) / d}
}
```

sideOf.h

Description: Returns where p is as seen from s towards e. 1/0/-1 ⇔ left/on line/right. If the optional argument eps is given 0 is returned if p is within distance eps from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.

Usage: bool left = side_of(p1,p2,q)==1;

"Point.h" 9e71fb, 9 lines

```
template<class P>
int side_of(P s, P e, P p) { return sgn(s.cross(e, p)); }

template<class P>
int side_of(const P& s, const P& e, const P& p, double eps) {
  auto a = (e - s).cross(p - s);
  double l = (e - s).dist() * eps;
  return (a > l) - (a < -l);
}
```

OnSegment.h

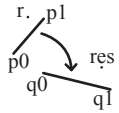
Description: Returns true iff p lies on the line segment from s to e. Use (seg_dist(s,e,p) < epsilon) instead when using Point<double>.

"Point.h" 0c1f75, 3 lines

```
template<class P> bool on_segment(P s, P e, P p) {
  return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

linearTransformation.h

Description:
Apply the linear transformation (translation, rotation and scaling) which takes line p0-p1 to line q0-q1 to point r.



"Point.h" 8e73be, 6 lines

```
typedef Point<double> P;
P linear_transformation(const P &p0, const P &p1,
    const P &q0, const P &q1, const P &r) {
  P dp = p1 - p0, dq = q1 - q0, num(dp.cross(dq), dp.dot(dq));
  return q0 + P((r - p0).cross(num), (r - p0).dot(num)) / dp.dist2();
}
```

LineProjectionReflection.h

Description: Projects point p onto line ab. Set refl=true to get reflection of point p across line ab instead. The wrong point will be returned if P is an integer point and the desired point doesn't have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow.

"Point.h" da84d5, 5 lines

```
template<class P>
P proj(P a, P b, P p, bool refl = false) {
  P v = b - a;
  return p - v.perp() * (1 + refl) * v.cross(p - a) / v.dist2();
}
```

Angle.h

Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.

Usage: vector<Angle> v = {w[0], w[0].t360() ...}; // sorted
int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }
// sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i

0f0602, 35 lines

```
struct Angle {
  int x, y;
  int t;
  Angle(int x, int y, int t = 0) : x(x), y(y), t(t) {}
  Angle operator-(Angle b) const { return {x - b.x, y - b.y, t}; }
  int half() const {
    assert(x || y);
    return y < 0 || (y == 0 && x < 0);
  }
  Angle t90() const { return {-y, x, t + (half() && x >= 0)}; }
  Angle t180() const { return {-x, -y, t + half()}; }
  Angle t360() const { return {x, y, t + 1}; }
};
bool operator<(Angle a, Angle b) {
  // add a.dist2() and b.dist2() to also compare distances
  return make_tuple(a.t, a.half(), a.y * (ll) b.x) <
```

```
      make_tuple(b.t, b.half(), a.x * (ll) b.y);
}

// Given two points, this calculates the smallest angle between
// them, i.e., the angle that covers the defined line segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
  if (b < a) swap(a, b);
  return (b < a.t180() ?
      make_pair(a, b) : make_pair(b, a.t360()));
}
Angle operator+(Angle a, Angle b) { // point a + vector b
  Angle r(a.x + b.x, a.y + b.y, a.t);
  if (a.t180() < r) r.t--;
  return r.t180() < a ? r.t360() : r;
}
Angle angleDiff(Angle a, Angle b) { // angle b - angle a
  int tu = b.t - a.t; a.t = b.t;
  return {a.x * b.x + a.y * b.y, a.x * b.y - a.y * b.x, tu - (b < a)};
}
```

8.5 Circles

CircleIntersection.h

Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.

"Point.h" b70280, 11 lines

```
using P = Point<db>;
bool circle_inter(P a, P b, db r1, db r2, pair<P, P> *out) {
  if (a == b) { assert(r1 != r2); return false; }
  P vec = b - a;
  db d2 = vec.dist2(), sum = r1 + r2, dif = r1 - r2,
       p = (d2 + r1 * r1 - r2 * r2) / (d2 * 2), h2 = r1 * r1 - p * p *
            d2;
  if (sum * sum < d2 || dif * dif > d2) return false;
  P mid = a + vec * p, per = vec.perp() * sqrt(fmax(0, h2) / d2);
  *out = {mid + per, mid - per};
  return true;
}
```

CircleTangents.h

Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0.

"Point.h" b0153d, 13 lines

```
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
  P d = c2 - c1;
  double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
  if (d2 == 0 || h2 < 0)  return {};
  vector<pair<P, P>> out;
  for (double sign : {-1, 1}) {
    P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
    out.push_back({c1 + v * r1, c2 + v * r2});
  }
  if (h2 == 0) out.pop_back();
  return out;
}
```

CircleLine.h

Description: Finds the intersection between a circle and a line. Returns a vector of either 0, 1, or 2 intersection points. P is intended to be Point<double>.

"Point.h" afd5f9, 9 lines

```
template<class P>
vector<P> circle_line(P c, double r, P a, P b) {
  P ab = b - a, p = a + ab * (c - a).dot(ab) / ab.dist2();
  double s = a.cross(b, c), h2 = r * r - s * s / ab.dist2();
  if (h2 < 0) return {};
  if (h2 == 0) return {p};
  P h = ab.unit() * sqrt(h2);
  return {p - h, p + h};
}
```

CirclePolygonIntersection.h

Description: Returns the area of the intersection of a circle with a ccw polygon.

Time: $\mathcal{O}(n)$

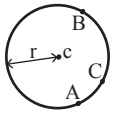
"../../content/geometry/Point.h" e876aa, 19 lines

```
typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
  auto tri = [&](P p, P q) {
    auto r2 = r * r / 2;
    P d = q - p;
    auto a = d.dot(p) / d.dist2(), b = (p.dist2() - r * r) / d.dist2()
        ;
    auto det = a * a - b;
    if (det <= 0) return arg(p, q) * r2;
    auto s = max(0., -a - sqrt(det)), t = min(1., -a + sqrt(det));
    if (t < 0 || 1 <= s) return arg(p, q) * r2;
    P u = p + d * s, v = q + d * (t - 1);
    return arg(p, u) * r2 + u.cross(v) / 2 + arg(v, q) * r2;
  };
  auto sum = 0.0;
  FOR (i, size(ps))
    sum += tri(ps[i] - c, ps[(i + 1) % size(ps)] - c);
  return sum;
}
```

circumcircle.h

Description:

The circumcirle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A, B and C and ccCenter returns the center of the same circle.



"Point.h" 30a12d, 9 lines

```
typedef Point<double> P;
double cc_radius(const P& A, const P& B, const P& C) {
  return (B - A).dist() * (C - B).dist() * (A - C).dist() /
      abs((B - A).cross(C - A)) / 2;
}
P cc_center(const P& A, const P& B, const P& C) {
  P b = C - A, c = B - A;
  return A + (b * c.dist2() - c * b.dist2()).perp() / b.cross(c) / 2;
}
```

MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points.

Time: expected $\mathcal{O}(n)$

"circumcircle.h" 256373, 17 lines

```
pair<P, double> mec(vector<P> ps) {
  shuffle(all(ps), mt19937(time(0)));
  P o = ps[0];
  double r = 0, EPS = 1 + 1e-8;
  FOR (i, size(ps)) if ((o - ps[i]).dist() > r * EPS) {
    o = ps[i], r = 0;
    FOR (j, i) if ((o - ps[j]).dist() > r * EPS) {
      o = (ps[i] + ps[j]) / 2;
      r = (o - ps[i]).dist();
      FOR (k, j) if ((o - ps[k]).dist() > r * EPS) {
        o = cc_center(ps[i], ps[j], ps[k]);
        r = (o - ps[i]).dist();
      }
    }
  }
  return {o, r};
}
```

8.6 Polygons

InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.

Usage: `vector<P> v = {P{4,4}, P{1,2}, P{2,1}};`
`bool in = in_polygon(v, P{3, 3}, false);`

Time: $\mathcal{O}(n)$

"Point.h", "OnSegment.h", "SegmentDistance.h" b915a1, 11 lines

```
template<class P>
bool in_polygon(vector<P> &p, P a, bool strict = true) {
  int cnt = 0, n = sz(p);
  FOR (i, n) {
    P q = p[(i + 1) % n];
    if (on_segment(p[i], q, a)) return !strict;
    //or: if (segDist(p[i], q, a) <= eps) return !strict;
    cnt ^= ((a.y < p[i].y) - (a.y < q.y)) * a.cross(p[i], q) > 0;
  }
  return cnt;
}
```

PolygonArea.h

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

"Point.h" 93542d, 6 lines

```
template<class T>
T polygon_area(vt<Point<T>>& v) {
  T a = v.back().cross(v[0]);
  FOR (i, size(v) - 1) a += v[i].cross(v[i + 1]);
  return a;
}
```

PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time: $\mathcal{O}(n)$

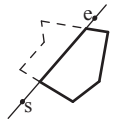
"Point.h" ce7a6a, 9 lines

```
typedef Point<db> P;
P polygon_center(const vector<P>& v) {
  P res(0, 0); db a = 0;
  for (int i = 0, j = size(v) - 1; i < size(v); j = i++) {
    res = res + (v[i] + v[j]) * v[j].cross(v[i]);
    a += v[j].cross(v[i]);
  }
  return res / a / 3;
}
```

PolygonCut.h

Description:

Returns a vector with the vertices of a polygon with everything to the left of the line going from s to e cut away.



Usage: `vector<P> p = ...;`
`p = polygonCut(p, P(0,0), P(1,0));`

"Point.h" d8e942, 13 lines

```
typedef Point<db> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
  vector<P> res;
  FOR (i, size(poly)) {
    P cur = poly[i], prev = i ? poly[i - 1] : poly.back();
    auto a = s.cross(e, cur), b = s.cross(e, prev);
```

```
    if ((a < 0) != (b < 0))
      res.push_back(cur + (prev - cur) * (a / (a - b)));
    if (a < 0)
      res.push_back(cur);
  }
  return res;
}
```

PolygonUnion.h

Description: Calculates the area of the union of n polygons (not necessarily convex). The points within each polygon must be given in CCW order. (Epsilon checks may optionally be added to sideOf/sgn, but shouldn't be needed.)
Time: $\mathcal{O}(N^2)$, where N is the total number of points

"Point.h", "sideOf.h" 8e652c, 33 lines

```
using P = Point<db>;
db rat(P a, P b) { return sgn(b.x) ? a.x / b.x : a.y / b.y; }
db poly_union(vt<vt<P>>& poly) {
  db ret = 0;
  FOR (i, size(poly)) FOR (v, size(poly[i])) {
    P A = poly[i][v], B = poly[i][(v + 1) % size(poly[i])];
    vt<pair<db, int>> segs = {{0, 0}, {1, 0}};
    FOR (j, size(poly)) if (i != j) {
      FOR (u, size(poly[j])) {
        P C = poly[j][u], D = poly[j][(u + 1) % size(poly[j])];
        int sc = side_of(A, B, C), sd = side_of(A, B, D);
        if (sc != sd) {
          db sa = C.cross(D, A), sb = C.cross(D, B);
          if (min(sc, sd) < 0)
            segs.emplace_back(sa / (sa - sb), sgn(sc - sd));
        } else if (!sc && !sd && j < i && sgn((B - A).dot(D-C)) > 0){
          segs.emplace_back(rat(C - A, B - A), 1);
          segs.emplace_back(rat(D - A, B - A), -1);
        }
      }
    }
    sort(all(segs));
    for (auto& s : segs) s.first = min(max(s.first, 0.0), 1.0);
    db sum = 0;
    int cnt = segs[0].second;
    FOR (j, 1, size(segs)) {
      if (!cnt) sum += segs[j].first - segs[j - 1].first;
      cnt += segs[j].second;
    }
    ret += A.cross(B) * sum;
  }
  return ret / 2;
}
```

ConvexHull.h

Description:
Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.

Time: $\mathcal{O}(n \log n)$

"Point.h" 5e9915, 12 lines

```
vt<P> convex_hull(vt<P> pts) {
  if (size(pts) <= 1) return pts;
  sort(all(pts));
  vector<P> h(size(pts)+1);
  int s = 0, t = 0;
  for (int it = 2; it--; s = --t, reverse(all(pts)))
    for (P p : pts) {
      while (t >= s + 2 && h[t - 2].cross(h[t - 1], p) <= 0) t--;
      h[t++] = p;
    }
  return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}
```

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).
Time: $\mathcal{O}(n)$

"Point.h" 8279d4, 12 lines

```
typedef Point<ll> P;
array<P, 2> hull_diameter(vector<P> s) {
  int n = size(s), j = n < 2 ? 0 : 1;
  pair<ll, array<P, 2>> res({0, {s[0], s[0]}});
  FOR (i, j)
    for (;; j = (j + 1) % n) {
      res = max(res, {(s[i] - s[j]).dist2(), {s[i], s[j]}});
      if ((s[(j + 1) % n] - s[j]).cross(s[i + 1] - s[i]) >= 0)
        break;
    }
  return res.second;
}
```

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.
Time: $\mathcal{O}(\log N)$

"Point.h", "sideOf.h", "OnSegment.h" c366a3, 13 lines

```
using P = Point<ll>;
bool in_hull(const vector<P>& l, P p, bool strict = true) {
  int a = 1, b = size(l) - 1, r = !strict;
  if (size(l) < 3) return r && on_segment(l[0], l.back(), p);
  if (side_of(l[0], l[a], l[b]) > 0) swap(a, b);
  if (side_of(l[0], l[a], p) >= r || side_of(l[0], l[b], p) <= -r)
    return false;
  while (abs(a - b) > 1) {
    int c = (a + b) / 2;
    (side_of(l[0], l[c], p) > 0 ? b : a) = c;
  }
  return sgn(l[a].cross(l[b], p)) < r;
}
```

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: • $(-1,-1)$ if no collision, • $(i,-1)$ if touching the corner i, • (i,i) if along side $(i,i+1)$, • (i,j) if crossing sides $(i,i+1)$ and $(j,j+1)$. In the last case, if a corner i is crossed, this is treated as happening on side $(i,i+1)$. The points are returned in the same order as the line hits the polygon. `extrVertex` returns the point of a hull with the max projection onto a line.
Time: $\mathcal{O}(\log n)$

"Point.h" bf84fc, 39 lines

```
#define cmp(i,j) sgn(dir.perp().cross(poly[(i) % n] - poly[(j) % n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
  int n = size(poly), lo = 0, hi = n;
  if (extr(0)) return 0;
  while (lo + 1 < hi) {
    int m = (lo + hi) / 2;
    if (extr(m)) return m;
    int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
    (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
  }
  return lo;
}

#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
  int endA = extrVertex(poly, (a - b).perp());
  int endB = extrVertex(poly, (b - a).perp());
  if (cmpL(endA) < 0 || cmpL(endB) > 0)
    return {-1, -1};
  array<int, 2> res;
  FOR (i, 2) {
    int lo = endB, hi = endA, n = size(poly);
    while ((lo + 1) % n != hi) {
      int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
      (cmpL(m) == cmpL(endB) ? lo : hi) = m;
    }
    res[i] = (lo + !cmpL(hi)) % n;
    swap(endA, endB);
  }
  if (res[0] == res[1]) return {res[0], -1};
  if (!cmpL(res[0]) && !cmpL(res[1]))
    switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
      case 0: return {res[0], res[0]};
      case 2: return {res[1], res[1]};
    }
  return res;
}
```

MinkowskiSum.h

Description: Minkowski sum of two convex polygons given in CCW order.
Time: $\mathcal{O}(N)$

e5d935, 29 lines

```
vP minkowski_sum(vP a, vP b) {
  if (sz(a) > sz(b)) swap(a, b);
  if (!sz(a)) return {};
  if (sz(a) == 1) {
    each(t, b) t += a.ft;
    return b;
  }
  rotate(begin(a), min_element(all(a)), end(a));
  rotate(begin(b), min_element(all(b)), end(b));
  a.pb(a[0]), a.pb(a[1]);
  b.pb(b[0]), b.pb(b[1]);
  vP result;
  int i = 0, j = 0;
  while (i < sz(a)-2 || j < sz(b)-2) {
    result.pb(a[i]+b[j]);
    T crs = cross(a[i+1]-a[i],b[j+1]-b[j]);
    i += (crs >= 0);
    j += (crs <= 0);
  }
  return result;
}

T diameter2(vP p) { // example application: squared diameter
  vP a = hull(p);
  vP b = a; each(t, b) t *= -1;
  vP c = minkowski_sum(a, b);
  T ret = 0; each(t, c) ckmax(ret, abs2(t));
  return ret;
}
```

8.7 Misc. Point Set Problems

ClosestPair.h

Description: Finds the closest pair of points.
Time: $\mathcal{O}(n \log n)$

"Point.h" be22ff, 16 lines

```
pair<P, P> closest(vector<P> v) {
  assert(size(v) > 1);
  set<P> S;
  sort(all(v), [](P a, P b) { return a.y < b.y; });
  pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
  int j = 0;
  for (P p : v) {
    P d{1 + (ll)sqrt(ret.first), 0};
```

```
    while (v[j].y <= p.y - d.x) S.erase(v[j++]);
    auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
    for (; lo != hi; ++lo)
      ret = min(ret, {(*lo - p).dist2(), {*lo, p}});
    S.insert(p);
  }
  return ret.second;
}
```

ManhattanMST.h

Description: Given N points, returns up to 4*N edges, which are guaranteed to contain a minimum spanning tree for the graph with edge weights w(p, q) = |p.x - q.x| + |p.y - q.y|. Edges are in the form (distance, src, dst). Use a standard MST algorithm on the result to find the final MST.
Time: $\mathcal{O}(N \log N)$

"Point.h" c36e89, 23 lines

```
typedef Point<int> P;
vt<array<int, 3>> manhattanMST(vt<P> ps) {
  vi id(size(ps));
  iota(all(id), 0);
  vector<array<int, 3>> edges;
  FOR (k, 4) {
    sort(all(id), [&](int i, int j) {
       return (ps[i]-ps[j]).x < (ps[j]-ps[i]).y;});
    map<int, int> sweep;
    for (int i : id) {
      for (auto it = sweep.lower_bound(-ps[i].y);
            it != sweep.end(); sweep.erase(it++)) {
        int j = it->second;
        P d = ps[i] - ps[j];
        if (d.y > d.x) break;
        edges.push_back({d.y + d.x, i, j});
      }
      sweep[-ps[i].y] = i;
    }
    for (P& p : ps) if (k & 1) p.x = -p.x; else swap(p.x, p.y);
  }
  return edges;
}
```

kdTree.h

Description: KD-tree (2d)

"Point.h" ad9b75, 53 lines

```
using P = array<int, 2>;
struct Node {
  #define _sq(x) (x) * (x)
  P lo, hi;
  struct Node *lc, *rc;

  ll dist2(const P &a, const P &b) const {
    return 1ll * _sq(a[0] - b[0]) + 1ll * _sq(a[1] - b[1]);
  }

  ll dist2(P &p) {
    #define _loc(i) (p[i] < lo[i] ? lo[i] : (p[i] > hi[i] ? hi[i] : p[
        i]))
    return dist2(p, {_loc(0), _loc(1)});
    // ll res = 0;
    // FOR (i, 2) {
    //   ll tmp = (p[i] < lo[i] ? lo[i] - p[i] : 0) + (hi[i] < p[i] ?
        p[i] - hi[i] : 0);
    //   res += tmp * tmp;
    // }
    // return res;
  }

  template<class ptr>
  Node (ptr l, ptr r, int d) : lc(0), rc(0) {
    lo = {inf, inf}, hi = {-inf, -inf};
    for (ptr p = l; p < r; p++) {
      FOR (i, 2) lo[i] = min(lo[i], (*p)[i]), hi[i] = max(hi[i], (*p)[
          i]);
    }
    if (r - l == 1) return;
    ptr m = l + (r - l) / 2;
    nth_element(l, m, r, [&] (auto a, auto b) { return a[d] < b[d]; })
        ;
    lc = new Node(l, m, d ^ 1);
    rc = new Node(m, r, d ^ 1);
  }

  void search(P p, ll &best) {
    if (lc) { // rc will also exist
      ll dl = lc->dist2(p), dr = rc->dist2(p);
      if (dl > dr) swap(lc, rc), dr = dl;
      lc->search(p, best);
      if (dr < best) rc->search(p, best);
    } else best = min(best, dist2(p, lo));
  }

  // fill pq with k infinities for nearest k points
  void search(P p, priority_queue<ll> &pq) {
    if (lc) {
      ll dl = lc->dist2(p), dr = rc->dist2(p);
      if (dl > dr) swap(lc, rc), dr = dl;
      lc->search(p, pq);
      if (dr < pq.top()) rc->search(p, pq);
    } else pq.push(dist2(p, lo)), pq.pop();
  }
};
```

FastDelaunay.h

Description: Fast Delaunay triangulation. Each circumcircle contains none of the input points. There must be no duplicate points. If all points are on a line, no triangles will be returned. Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in order {t[0][0], t[0][1], t[0][2], t[1][0], ...}, all counter-clockwise.
Time: $\mathcal{O}(n \log n)$

"Point.h" eefdf5, 88 lines

```
typedef Point<ll> P;
typedef struct Quad* Q;
typedef __int128_t lll; // (can be ll if coords are < 2e4)
P arb(LLONG_MAX,LLONG_MAX); // not equal to any other point

struct Quad {
  Q rot, o; P p = arb; bool mark;
  P& F() { return r()->p; }
  Q& r() { return rot->rot; }
  Q prev() { return rot->o->rot; }
  Q next() { return r()->prev(); }
} *H;

bool circ(P p, P a, P b, P c) { // is p in the circumcircle?
  lll p2 = p.dist2(), A = a.dist2()-p2,
    B = b.dist2()-p2, C = c.dist2()-p2;
  return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*B > 0;
}
Q makeEdge(P orig, P dest) {
  Q r = H ? H : new Quad{new Quad{new Quad{new Quad{0}}}};
  H = r->o; r->r()->r() = r;
  rep(i,0,4) r = r->rot, r->p = arb, r->o = i & 1 ? r : r->r();
  r->p = orig; r->F() = dest;
  return r;
}
void splice(Q a, Q b) {
  swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
}
Q connect(Q a, Q b) {
  Q q = makeEdge(a->F(), b->p);
  splice(q, a->next());
  splice(q->r(), b);
  return q;
}

pair<Q,Q> rec(const vector<P>& s) {
  if (sz(s) <= 3) {
    Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.back());
    if (sz(s) == 2) return { a, a->r() };
    splice(a->r(), b);
    auto side = s[0].cross(s[1], s[2]);
    Q c = side ? connect(b, a) : 0;
    return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
  }

#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
  Q A, B, ra, rb;
  int half = sz(s) / 2;
  tie(ra, A) = rec({all(s) - half});
  tie(B, rb) = rec({sz(s) - half + all(s)});
  while ((B->p.cross(H(A)) < 0 && (A = A->next())) ||
       (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
  Q base = connect(B->r(), A);
  if (A->p == ra->p) ra = base->r();
  if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (circ(e->dir->F(), H(base), e->F())) { \
      Q t = e->dir; \
      splice(e, e->prev()); \
      splice(e->r(), e->r()->prev()); \
      e->o = H; H = e; e = t; \
    }
  for (;;) {
    DEL(LC, base->r(), o);  DEL(RC, base, prev());
    if (!valid(LC) && !valid(RC)) break;
    if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
      base = connect(RC, base->r());
    else
      base = connect(base->r(), LC->r());
  }
  return { ra, rb };
}

vector<P> triangulate(vector<P> pts) {
  sort(all(pts));  assert(unique(all(pts)) == pts.end());
  if (sz(pts) < 2) return {};
  Q e = rec(pts).first;
  vector<Q> q = {e};
  int qi = 0;
  while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c->p); \
  q.push_back(c->r()); c = c->next(); } while (c != e); }
  ADD; pts.clear();
  while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
  return pts;
}
```

AllPointPairs.h

Description: considers points sorted with respect to every direction

7a2fce, 32 lines

```
int n; cin >> n;
vt<P> pts(n);
for (auto &[x, y] : pts) cin >> x >> y;

vi ord(n), loc(n);
```

```
FOR (i, n) ord[i] = i;
sort(all(ord), [&] (int i, int j) { return pts[i].x < pts[j].x; });
FOR (i, n) loc[ord[i]] = i;

// when the dot vector reaches P, x should be greater than y
vt<tuple<P, int, int>> evs;
FOR (x, n) {
  FOR (y, n) {
    if (x == y) continue;
    // consider only adding .half() == 0 for better runtime
    evs.pb({(pts[y] - pts[x]).perp(), x, y});
  }
}
sort(all(evs));
ll mn = INF, mx = -INF;

for (auto [dir, x, y] : evs) {
  int &i = loc[x], &j = loc[y];
  if (i < j) swap(ord[i], ord[j]), swap(i, j);
  // x is "greater" than y with respect to dir (ie. i > j) and x is on
        the left
  if (i - j > 1) mn = 0;
  if (i + 1 < n) mn = min(mn, pts[x].cross(pts[y], pts[ord[i + 1]]));
  if (j - 1 >= 0) mn = min(mn, -pts[x].cross(pts[y], pts[ord[j - 1]]))
      ;

  mx = max(mx, abs(pts[x].cross(pts[y], pts[ord[0]])));
  mx = max(mx, abs(pts[x].cross(pts[y], pts[ord[n - 1]])));
}
```

Strings (9)

KMP.h

Description: pi[x] computes the length of the longest prefix of s that ends at x, other than s[0...x] itself (abacaba -> 0010123). Can be used to find all occurrences of a string.
Time: $\mathcal{O}(n)$

d4375c, 16 lines

```
vi pi(const string& s) {
  vi p(sz(s));
  rep(i,1,sz(s)) {
    int g = p[i-1];
    while (g && s[i] != s[g]) g = p[g-1];
    p[i] = g + (s[i] == s[g]);
  }
  return p;
}

vi match(const string& s, const string& pat) {
  vi p = pi(pat + '\0' + s), res;
  rep(i,sz(p)-sz(s),sz(p))
    if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
  return res;
}
```

Zfunc.h

Description: z[i] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)
Time: $\mathcal{O}(n)$

98e47b, 12 lines

```
vi Z(const string& S) {
  vi z(size(S));
  int l = -1, r = -1;
  FOR (i, 1, size(S)) {
    z[i] = i >= r ? 0 : min(r - i, z[i - l]);
    while (i + z[i] < size(S) && S[i + z[i]] == S[z[i]])
      z[i]++;
    if (i + z[i] > r)
      l = i, r = i + z[i];
  }
  return z;
}
```

BigMod.h

Description: $2^{64} - 1$ mod int

ca9e8e, 18 lines

```
// Arithmetic mod 2^64-1. 2x slower than mod 2^64 and more
// code, but works on evil test data (e.g. Thue-Morse, where
// ABBA... and BAAB... of length 2^10 hash the same mod 2^64).
// "typedef ull H;" instead if you think test data is random,
// or work mod 10^9+7 if the Birthday paradox is not a problem.
using ull = unsigned long long;
struct H {
  ull x; H(ull x = 0) : x(x) {}
  H operator+(H o) { return x + o.x + (x + o.x < x); }
  H operator-(H o) { return *this + ~o.x; }
  H operator*(H o) { auto m = (__uint128_t) x * o.x;
    return H((ull) m) + (ull)(m >> 64); }
  ull get() const { return x + !~x; }
  #define op(o) bool operator o (H oth) const { return get() o oth.get
      (); }
  op(==) op(<)
};

static const H C = (ll) 1e11 + 3; // (order ~ 3e9; random also ok)
```

Hashing.h

Description: Self-explanatory methods for string hashing. Skip the stuff that starts with r if you don't care about reverse hashes etc.

"BigMod.h" 35b1ca, 39 lines

```
struct HashInterval {
  vt<H> ha, pw, rha;
  template<class T>
  HashInterval(T& str) : ha(size(str) + 1), pw(ha), rha(ha) {
    pw[0] = 1;
    FOR (i, size(str)) {
      ha[i + 1] = ha[i] * C + str[i] + 1;
      pw[i + 1] = pw[i] * C;
    }
    ROF (i, 0, size(str)) rha[i] = rha[i + 1] * C + str[i] + 1;
  }
  H hash_interval(int l, int r) { // hash [l, r)
    return ha[r] - ha[l] * pw[r - l];
  }
  H rhash_interval(int l, int r) { // hash [l, rf) from right to left
    return rha[l] - rha[r] * pw[r - l];
  }
};

// get all hashes of length <len>
template<class T>
vt<H> get_hashes(T& str, int length) {
  if (size(str) < length) return {};
  H h = 0, pw = 1;
  FOR (i, length) h = h * C + str[i] + 1, pw = pw * C;
  vector<H> ret = {h};
  FOR (i, length, size(str)) {
    ret.push_back(h = h * C + str[i] + 1
      - pw * (str[i - length] + 1));
  }
  return ret;
}

template<class T>
H hash_string(T& s) {
  H h = 1;
  for (auto c : s) h = h * C + c + 1;
  return h;
}
```

Manacher.h

Description: For each position in a string, computes p[0][i] = half length of longest even palindrome around pos i, p[1][i] = longest odd (half rounded down).
Time: $\mathcal{O}(N)$

1c24a1, 13 lines

```
array<vi, 2> manacher(const string &s) {
  int n = size(s);
  array<vi, 2> p = {vi(n + 1), vi(n)};
  FOR (z, 2) for (int i = 0, l = 0, r = 0; i < n; i++) {
    int t = r - i + !z;
    if (i < r) p[z][i] = min(t, p[z][l + t]);
    int L = i - p[z][i], R = i + p[z][i] - !z;
    while (L >= 1 && R + 1 < n && s[L - 1] == s[R + 1])
      p[z][i]++, L--, R++;
    if (R > r) l = L, r = R;
  }
  return p;
}
```

MinRotation.h

Description: Finds the lexicographically smallest rotation of a string.
Usage: rotate(v.begin(), v.begin() + minRotation(v), v.end());
Time: $\mathcal{O}(N)$

9bd725, 8 lines

```
int minRotation(string s) {
  int a = 0, N = size(s); s += s;
  FOR (b, N) FOR (k, N) {
    if (a + k == b || s[a + k] < s[b + k]) { b += max(0, k - 1); break
        ; }
    if (s[a + k] > s[b + k]) { a = b; break; }
  }
  return a;
}
```

SuffixArray.h

Description: Builds suffix array for a string. sa[i] is the starting index of the suffix which is i'th in the sorted suffix array. The returned vector is of size $n+1$, and sa[0] = n. The lcp array contains longest common prefixes for neighbouring strings in the suffix array: lcp[i] = lcp(sa[i], sa[i-1]), lcp[0] = 0. The input string must not contain any nul chars.
Time: $\mathcal{O}(N \log N)$

a982c3, 22 lines

```
struct SuffixArray {
  vi sa, lcp;
  SuffixArray(string s, int lim = 256) { // or vector<int>
    s.push_back(0); int n = size(s), k = 0, a, b;
    vi x(all(s)), y(n), ws(max(n, lim));
    sa = lcp = y, iota(all(sa), 0);
    for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
      p = j, iota(all(y), n - j);
      FOR (i, n) if (sa[i] >= j) y[p++] = sa[i] - j;
      fill(all(ws), 0);
      FOR (i, n) ws[x[i]]++;
      FOR (i, 1, lim) ws[i] += ws[i - 1];
      for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
      swap(x, y), p = 1, x[sa[0]] = 0;
      FOR (i, 1, n) a = sa[i - 1], b = sa[i], x[b] =
        (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
    }
    for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
      for (k && k--, j = sa[x[i] - 1];
          s[i + k] == s[j + k]; k++);
  }
};
```

SuffixTree.h

Description: Ukkonen's algorithm for online suffix tree construction. Each node contains indices [l, r) into the string, and a list of child nodes. Suffixes are given by traversals of this tree, joining [l, r) substrings. The root is 0 (has l = -1, r = 0), non-existent children are -1. To get a complete tree, append a dummy symbol – otherwise it may contain an incomplete path (still useful for substring matching, though).

Time: $\mathcal{O}(26N)$

aae0b8, 50 lines

```
struct SuffixTree {
  enum { N = 200010, ALPHA = 26 }; // N ~ 2*maxlen+10
  int toi(char c) { return c - 'a'; }
  string a; // v = cur node, q = cur position
  int t[N][ALPHA],l[N],r[N],p[N],s[N],v=0,q=0,m=2;

  void ukkadd(int i, int c) { suff:
    if (r[v]<=q) {
      if (t[v][c]==-1) { t[v][c]=m;  l[m]=i;
        p[m++]=v; v=s[v]; q=r[v];  goto suff; }
      v=t[v][c]; q=l[v];
    }
    if (q==-1 || c==toi(a[q])) q++; else {
      l[m+1]=i;  p[m+1]=m;  l[m]=l[v];  r[m]=q;
      p[m]=p[v];  t[m][c]=m+1;  t[m][toi(a[q])]=v;
      l[v]=q;  p[v]=m;  t[p[m]][toi(a[l[m]])]=m;
      v=s[p[m]];  q=l[m];
      while (q<r[m]) { v=t[v][toi(a[q])];  q+=r[v]-l[v]; }
      if (q==r[m])  s[m]=v;  else s[m]=m+2;
      q=r[v]-(q-r[m]);  m+=2;  goto suff;
    }
  }

  SuffixTree(string a) : a(a) {
    fill(r,r+N,sz(a));
    memset(s, 0, sizeof s);
    memset(t, -1, sizeof t);
    fill(t[1],t[1]+ALPHA,0);
    s[0] = 1; l[0] = l[1] = -1; r[0] = r[1] = p[0] = p[1] = 0;
    rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
  }

  // example: find longest common substring (uses ALPHA = 28)
  pii best;
  int lcs(int node, int i1, int i2, int olen) {
    if (l[node] <= i1 && i1 < r[node]) return 1;
    if (l[node] <= i2 && i2 < r[node]) return 2;
    int mask = 0, len = node ? olen + (r[node] - l[node]) : 0;
    rep(c,0,ALPHA) if (t[node][c] != -1)
      mask |= lcs(t[node][c], i1, i2, len);
    if (mask == 3)
      best = max(best, {len, r[node] - len});
    return mask;
  }
  static pii LCS(string s, string t) {
    SuffixTree st(s + (char)('z' + 1) + t + (char)('z' + 2));
    st.lcs(0, sz(s), sz(s) + 1 + sz(t), 0);
    return st.best;
  }
};
```

Heuristics (10)

10.1 Tips

10.1.1 Simulated Annealing

Default temperature schedule for SA: $t = t_0 * (\frac{t_n}{t_0})^{\text{time_passed}}$, time_passed $\in [0, 1]$ Acceptance function: $RNG() < exp((\text{cur} - \text{new})/t)$ A good schedule should have a slowly decreasing acceptance rate. Priorities:

- Prototype different transitions.
- Optimize eval function (= make it dynamic).
- Find a proper temp schedule. Consider starting with lower temperatures and increasing until it starts getting worse.

Things to do if its easy to get stuck in a local optimum:

- Use a very high temperature/acceptance rate
- Complex transitions
- Scale transitions with temperature (maybe?)
- (rarely) kicks
- (rarely) restarting temp schedule
- (rarely) multiple runs

Things to do when there are lots of invalid states:

- Add an additional scoring component based on number of collisions
- Do many short SA runs with high temp on partial subsolutions

If your eval is very fast, make sure RNG, exp and cur_time are not bottlenecks. When SA is bad:

- Very easy to get stuck in local optima
- Simple transitions are mixed with complex ones – have a separate acceptance function for each transition
- Eval is erratic during transitions

Use a good local tester: make sure it has multithreading and relative scoring. If you are considering whether its worth optimising speed, just run with higher time limit and see if the improvements are worth it. Removing bugs is #1 priority; results are meaningless when bugs are involved. If results do not align with intuition, investigate. Merge N-dimensional coordinates into 1D. Memory allocation is slow.

SimAnneal.h

Description: template for simulated annealing

<bits/extc++.h>, <sys/time.h> 88b123, 122 lines

```
#pragma GCC optimize("Ofast,omit-frame-pointer,inline,unroll-all-loops
    ")
#pragma GCC target("avx2")

using namespace std;
using ll = long long;

// include standard macros
const int inf = 1e9;
const ll INF = 1e18;
using vi = vt<int>;
using db = double;

__gnu_cxx::sfmt19937 mt(random_device{}());
int gen(int l, int r) { return uniform_int_distribution<int>(l, r)(mt)
    ; }
db next_double() { return uniform_real_distribution<db>(0, 1)(mt); }

double get_time() {timeval tv; gettimeofday(&tv, NULL); return tv.
    tv_sec + tv.tv_usec * 1e-6;}
double start_time = get_time();
double elapsed() {return get_time() - start_time;}

const db TIME_LIMIT = 0.95;
db t0 = 1e9, tn = 0.1, time_passed = 1e-9;
const db maximise_score = -1; // -1 if minimise

// data
int n;
vector<int> a;
// end data

using T = int;
struct State {
  T value;
  vi a;

  // static transition: consider optimising to dynamic
  void to_neighbour() {
    int i, j;
    switch(gen(0, 4)) {
      case 0:
        i = gen(0, n - 1);
        j = gen(0, n - 2);
        if (i == j) j++;
        swap(a[i], a[j]);
        break;
      case 1:
        a[gen(0, n - 1)] *= -1;
        break;
      case 2:
      case 3:
        i = gen(0, n);
        j = gen(0, n);
        if (i > j) swap(i, j);
        reverse(a.begin() + i, a.begin() + j);
        break;
      case 4:
        i = gen(0, n);
        j = gen(0, n);
        if (i > j) swap(i, j);
        FOR (k, i, j) a[k] *= -1;
        break;
    }
  }

  int calc_value() {
    T ans = 0, sum = 0;
    for (int x : a) sum += x, ans += abs(sum);
    return value = ans;
```

```
  }

  void print() {
    F0R (i, n) {
      if (a[i] > 0) cout << a[i] << " a\n";
      else cout << -a[i] << " f\n";
    }
    cout << value << '\n';
  }
};

State cur, best;

signed main() {
  cin.tie(0)->sync_with_stdio(0);

  cin >> n;
  a.resize(n);
  for (int &x : a) cin >> x;
  sort(all(a));
  F0R (i, n) if (a[i] % 4 == 1 || a[i] % 4 == 2) a[i] *= -1;
  cur.a = a;
  cur.calc_value();
  best = cur;
  t0 = cur.value * 2;

  if (n == 1) {
    cur.print();
    return 0;
  }

  int its = 0;
  db t;
  while (true) {
    if ((its & 511) == 0) {
      time_passed = elapsed() / TIME_LIMIT;
      if (time_passed > 1.0) break;
      t = t0 * pow(tn / t0, time_passed);
    }
    its++;

    State neighbour = cur;
    neighbour.to_neighbour();
    if ((neighbour.calc_value() - cur.value) * maximise_score >= 0
      || next_double() < exp(((neighbour.value - cur.value) *
          maximise_score) / t)) {
      cur = neighbour;
      if ((cur.value - best.value) * maximise_score > 0) {
        best = cur;
      }
    }
  }
  best.print();
}
```

Various (11)

11.1 Game Theory

Impartial games: Both players have the same set of moves from any position. Normal play: Last move wins

11.1.1 Grundy Numbers

For some position P:

$$G(P) = \text{mex}\{\, G(P') \mid P' \text{ is reachable from } P \,\}$$

Idea: $G(P) = 0$ iff P is losing.
Sprague-Grundy theorem: if you combine independent games $P_1, P_2, \ldots, P_k$:

$$G(\text{sum}) = G(P_1) \oplus G(P_2) \oplus \cdots \oplus G(P_k)$$

Green Hackenbush: A subtree can be replaced with a single line with length equal to xor of its subtrees. A cycle can be contracted into a single node with contracted edges sticking out.

11.2 Misc. algorithms

TernarySearch.h

Description: Find the smallest i in $[a, b]$ that maximizes $f(i)$, assuming that $f(a) < \ldots < f(i) \geq \cdots \geq f(b)$. To reverse which of the sides allows non-strict inequalities, change the < marked with (A) to <=, and reverse the loop at (B). To minimize f, change it to >, also at (B).
Usage: `int ind = ternSearch(0,n-1,[&](int i){return a[i];});`
Time: $\mathcal{O}(\log(b-a))$

9155b4, 11 lines

```
template<class F>
int ternSearch(int a, int b, F f) {
  assert(a <= b);
  while (b - a >= 5) {
    int mid = (a + b) / 2;
    if (f(mid) < f(mid+1)) a = mid; // (A)
    else b = mid+1;
  }
  rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B)
  return a;
}
```

11.3 Dynamic programming

KnuthDP.h

Description: When doing DP on intervals: $a[i][j] = \min_{i<k<j}(a[i][k] + a[k][j]) + f(i,j)$, where the (minimal) optimal k increases with both i and j, one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j-1]$ and $p[i+1][j]$. This is known as Knuth DP. Sufficient criteria for this are if $f(b,c) \leq f(a,d)$ and $f(a,c) + f(b,d) \leq f(a,d) + f(b,c)$ for all $a \leq b \leq c \leq d$. Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.
Time: $\mathcal{O}(N^2)$

DivideAndConquerDP.h

Description: Given $a[i] = \min_{lo(i) \leq k < hi(i)}(f(i,k))$ where the (minimal) optimal k increases with i, computes $a[i]$ for $i = L..R-1$.
Time: $\mathcal{O}((N + (hi - lo)) \log N)$

d38d2b, 18 lines

```
struct DP { // Modify at will:
  int lo(int ind) { return 0; }
  int hi(int ind) { return ind; }
  ll f(int ind, int k) { return dp[ind][k]; }
  void store(int ind, int k, ll v) { res[ind] = pii(k, v); }

  void rec(int L, int R, int LO, int HI) {
    if (L >= R) return;
    int mid = (L + R) >> 1;
    pair<ll, int> best(LLONG_MAX, LO);
    rep(k, max(LO,lo(mid)), min(HI,hi(mid)))
      best = min(best, make_pair(f(mid, k), k));
    store(mid, best.second, best.first);
    rec(L, mid, LO, best.second+1);
    rec(mid+1, R, best.second, HI);
  }
  void solve(int L, int R) { rec(L, R, INT_MIN, INT_MAX); }
};
```

11.4 Debugging tricks

- `signal(SIGSEGV, [](int) { _Exit(0); });` converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). `_GLIBCXX_DEBUG` failures generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- `feenableexcept(29);` kills the program on NaNs (`1`), 0-divs (`4`), infinities (`8`) and denormals (`16`).

LeakSanitizer.h

Description: tspmo

bc29d7, 4 lines

```
#ifdef __cplusplus
extern "C"
#endif
const char* __asan_default_options() { return "detect_leaks=0"; }
```

11.5 Optimization tricks

`__builtin_ia32_ldmxcsr(40896);` disables denormals (which make floats 20x slower near their minimum value).

11.5.1 Bit hacks

- `x & -x` is the least bit in `x`.
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of `m` (except `m` itself).
- `c = x & -x, r = x + c; (((r ^ x) >> 2) / c) | r` is the next number after `x` with the same number of bits set.
- `F0R (b, K) F0R (i, (1 << K))`
  `if (i & 1 << b) D[i] += D[i ^ (1 << b)];`
  computes all sums of subsets.

11.5.2 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize loops and optimizes floating points better.
- `#pragma GCC target ("avx2")` can double performance of vectorized code, but causes crashes on old machines.
- `#pragma GCC optimize ("O0", "trapv")` kills the program on integer overflows (but is really slow).

FastMod.h

Description: Compute $a\%b$ about 5 times faster than usual, where b is constant but not known at compile time. Returns a value congruent to $a \pmod b$ in the range $[0, 2b)$.

751a02, 8 lines

```
typedef unsigned long long ull;
struct FastMod {
  ull b, m;
  FastMod(ull b) : b(b), m(-1ULL / b) {}
  ull reduce(ull a) { // a % b + (0 or b)
    return a - (ull)((__uint128_t(m) * a) >> 64) * b;
  }
};
```

FastInput.h

Description: Read an integer from stdin. Usage requires your program to pipe in input from file.
Usage: `./a.out < input.txt`
Time: About 5x as fast as cin/scanf.

7b3c70, 17 lines

```
inline char gc() { // like getchar()
  static char buf[1 << 16];
  static size_t bc, be;
  if (bc >= be) {
    buf[0] = 0, bc = 0;
    be = fread(buf, 1, sizeof(buf), stdin);
  }
  return buf[bc++]; // returns 0 on EOF
}

int readInt() {
  int a, c;
  while ((a = gc()) < 40);
  if (a == '-') return -readInt();
  while ((c = gc()) >= 48) a = a * 10 + c - 480;
  return a - 48;
}
```

BumpAllocator.h

Description: When you need to dynamically allocate many objects and don't care about freeing them. "new X" otherwise has an overhead of something like 0.05us + 16 bytes per allocation.

745db2, 8 lines

```
// Either globally or in a single class:
static char buf[450 << 20];
void* operator new(size_t s) {
  static size_t i = sizeof buf;
  assert(s < i);
  return (void*)&buf[i -= s];
}
void operator delete(void*) {}
```

SmallPtr.h

Description: A 32-bit pointer that points into BumpAllocator memory.

"BumpAllocator.h" 2dd6c9, 10 lines

```
template<class T> struct ptr {
  unsigned ind;
  ptr(T* p = 0) : ind(p ? unsigned((char*)p - buf) : 0) {
    assert(ind < sizeof buf);
  }
  T& operator*() const { return *(T*)(buf + ind); }
  T* operator->() const { return &**this; }
  T& operator[](int a) const { return (&**this)[a]; }
  explicit operator bool() const { return ind; }
};
```

BumpAllocatorSTL.h

Description: BumpAllocator for STL containers.
Usage: `vector<vector<int, small<int>>> ed(N);`

bb66d4, 14 lines

```
char buf[450 << 20] alignas(16);
size_t buf_ind = sizeof buf;

template<class T> struct small {
  typedef T value_type;
  small() {}
  template<class U> small(const U&) {}
  T* allocate(size_t n) {
    buf_ind -= n * sizeof(T);
    buf_ind &= 0 - alignof(T);
    return (T*)(buf + buf_ind);
  }
  void deallocate(T*, size_t) {}
};
```

Unrolling.h

520e76, 5 lines

```
#define F {...; ++i;}
int i = from;
while (i&3 && i < to) F // for alignment, if needed
while (i + 4 <= to) { F F F F }
while (i < to) F
```

11.6 Other

GrayCode.h

Description: Gray codes

e87165, 10 lines

```
int g(int n) {
  return n ^ (n >> 1);
}

int rev_g (int g) {
  int n = 0;
  for (; g; g >>= 1)
  n ^= g;
  return n;
}
```

STL.h

Description: how to do certain things in stl

37dc93, 4 lines

```
auto cmp = [&] (T a, T b) { return a < b; }
set<T, decltype(cmp)> s(cmp);
map<T, int, decltype(cmp)> m(cmp);
priority_queue<T, vt<T>, decltype(cmp)> pq(cmp); // max
```

Python.py

Description: In case you need to python crutch

37 lines

```
# might not need encoding='utf-8'
# write text
with open('in', 'w', encoding='utf-8') as f:
  f.write("Hello\n")
  f.write("Line 2\n")

# append text
with open('notes.txt', 'a', encoding='utf-8') as f:
  f.write("Appended line\n")

# read text
with open('notes.txt', 'r', encoding='utf-8') as f:
  all_text = f.read()      # whole file as str
  # or iterate lines:
  f.seek(0)
  for line in f:
    print(line.strip())

from decimal import Decimal, getcontext, localcontext, ROUND_HALF_UP,
     ROUND_HALF_EVEN
from fractions import Fraction

Decimal('0.1')
Decimal('123.456')
Decimal(10)

f = Fraction(1, 3)
Decimal(f.numerator) / Decimal(f.denominator)
d = Decimal('1.23')
Decimal(d)         # returns a copy

ctx = getcontext()
print(ctx.prec)    # integer precision (significant digits)
print(ctx.rounding)  # default rounding mode

# set global precision (affects operations that follow)
getcontext().prec = 50
Decimal('2').sqrt()
```

11.7 Test Session

TestSession.h

Description: Things to test

294485, 41 lines

```
Keyboard layouts
bashrc
vimrc
template.cpp
Python
Printing
Sending clarifications
Hash script
-fsanitize doesn't randomly brick
gdb works
Keybinds don't cause computer to turn off
Check available documentation
Submit script
Whitespace/case insensitivity in output
Return values from main
Printing to stderr
Source code limit
Is it possible to submit and read from non-source files?
SIMD support on machine and judge
Benchmark local vs judge runtime
How long do array accesses take on machine resp. judge?
  ORAC: ~0.8s for N = 2e6 min segtree with Q = 2e6 loops of query +
       update
How long do small operations take on machine resp. judge?
Does clock() work?
Do WA on samples count towards penalty?
All judge errors:
  Accepted
  WA
  Presentation Error
  RTE (what results in RTE?)
  TLE
  Memory limit, both stack and heap
  Output limit
  Illegal function
  Compile error
  Compile time limit exceeded
  Compile memory limit exceeded
  Too late
  Listing all available binaries
    echo $PATH | tr ':' ' ' | xargs ls | grep -v /
    | sort | uniq | tr '\n' ' ' > path.txt
```